

AOS-CX 10.16.xxxx Monitoring Guide

5420, 6200 Switch Series



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Chapter 1

About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Applicable products

This document applies to the following products:

- HPE Aruba Networking 5420 Switch Series (S0U67A, S0U55A, S0U63A, S0U64A, S0U65A, S0U75A, S0U72A, S0U78A, S0U58A, S0U73A, S0U74A, S0U71A, S0U76A, S0U70A, S0U77A, S0U60A, S0U61A, S0U62A, S0U66A, S0U68A)

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Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <code><example-text></code> ■ <i>example-text</i> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (<code>< ></code>). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Convention	Usage
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 6400 and 5420 Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Specifies physical location of a module in the switch chassis.
 - Management modules are on the front of the switch in slots 1/1 and 1/2.
 - Line modules are on the front of the switch starting in slot 1/3.
- *port*: Physical number of a port on a line module.

For example, the logical interface **1/3/4** in software is associated with physical port 4 in slot 3 on member 1.

Confirming normal operation of the switch by reading LEDs

This task describes using the switch LEDs to confirm that the switch is operating normally.



For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the [Aruba Switch Documentation](#) section of the [Aruba Hardware Documentation and Translations Portal](#).

Procedure

1. Quick check: Verify that the chassis has power and there are no fault conditions.
On the front of the switch, verify that the states of the following LEDs are On Green:
 - Power
 - Health
2. Verify that the Health LEDs of all installed line modules are On Green.
3. Verify that the Health LEDs of all installed management modules are On Green.
4. Verify that the network ports are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a line module is in one of the following states:
 - On Green (normal operation)
 - Off (no line module installed)
 - b. On each line module, verify that each port LED is in one of the following states:
 - On Green, Half-Bright Green, or Flickering Green (normal operation)
 - Off (no cable connected or port off by default in config)
5. Verify that the power supplies are operating normally.
 - a. On the active management module, check the Status Front section. Verify that each LED that indicates a power supply is in one of the following states:
 - On Green (normal operation)
 - Off (no power supply installed)
 - b. On each power supply, verify that LEDs are in the following states:
 - Power LED: On Green
 - Fault LED: Off
6. Verify that the rear components are operating normally by checking the Status Rear section of the active management module:
 - a. Verify that the LEDs for the fabric modules are in one of the following states:
 - On Green (normal operation)
 - Off (component not installed)
 - b. Verify that the LEDs for the fan trays and fans are On Green.

7. Verify that the standby management module is ready to take over as the active management module. On the standby management module, verify the states of the following LEDs:
 - Health LED is On Green.
 - Management state standby (Stby) LED is On Green.

Detecting if the switch is not ready for a failover event

This task describes using the switch LEDs to detect if the switch is not ready for the loss of a fabric module or for a failover from the active management module to the standby management module.



Although you can detect power supply failures by viewing the LEDs, you must use software commands to determine if the power supply redundancy is sufficient to power the chassis if a power supply fails. For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the [Aruba Switch Documentation](#) section of the [Aruba Hardware Documentation and Translations Portal](#).

Procedure

1. Detect if the standby management module is shut down.
If the standby management module is shut down, the LED states are as follows:
 - The standby management module health LED is Off.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Off. For example, if the active management module is Management Module LED 5, Management Modules LED 6 is Off.
2. Detect if the standby management module is in a transient state. If the standby management module is booting, updating, or in another transient state, the LED states are as follows:
 - The standby management module health LED is Slow Flash Green when the service operating system is running or during an operating system update.
 - The standby management module Booting LED is Slow Flash Green when the AOS-CX operating system is booting.
 - The standby management state active (Actv) LED is Off.
 - The standby management state standby (Stby) LED is Off.
 - On the active management module in the Status Front Management Modules section, the LED for the standby management module is Slow Flash Green.
3. Detect if a fabric module is shut down or not present. If a fabric module is shut down or not present, the LED states are as follows:
 - On the active management module, in the Status Rear section, the LED for the fabric module is Off.
 - On the rear display module, the LED for the fabric module is Off.
 - On the fabric module, the health LED is Off. However, the fabric module is behind fan 1 and is not directly visible.

Finding faulted components using the switch LEDs

This task describes using the switch LEDs to find components that are in a fault condition.



All green LEDs—except for chassis power LEDs and the Usr1 LED—are off when the LED mode is set to Light Faults (The Usr1 LED of the LED Mode section of the active management module is On Green and the default behavior for the Usr1 LED is being used.). For complete information on LED behaviors for your AOS-CX switch, refer to the **Installation and Getting Started Guide** for that switch series, available for download from the [Aruba Switch Documentation](#) section of the [Aruba Hardware Documentation and Translations Portal](#).

Procedure

1. Find the switch that has the fault condition, which is indicated by a chassis health LED in the state of Slow Flash Orange.

The chassis health LED is located on the front of the switch and on the rear panel of the switch.

2. If you are at the back of the switch, on the rear panel, look for LEDs that are in the Slow Flash Orange state:

The Status Rear area has LEDs for power supplies, fabric modules, fan trays, and fans. The number on the LED represents the unit number of the component.

If the only LED in a state of Slow Flash Orange is the Chassis health LED, go to the front of the switch.

3. At the front of the switch, on the active management module, look for LEDs that are in the Slow Flash Orange state:
 - The Status Front area has LEDs for power supplies, line and fabric modules, and management modules. The number on the LED indicates the slot number of the component.
 - The Status Rear area has LEDs for fabric modules and fan trays, with a single LED for all the fans in the fan tray. The number on the LED represents the slot or bay number of the component.

4. Use the number indicated by the LED that is flashing to locate the slot that contains the faulted component.

The fabric modules are located behind the fan trays, and the fabric module number corresponds to the fan tray number.

5. At the front of the switch, on line modules, look for LEDs that are in the Slow Flash Orange state: Module LEDs and Port LEDs indicate faults if their states are Slow Flash Orange.

Chapter 3

IP Flow Information Export

IP Flow Information Export (IPFIX) is an embedded network flow analysis tool that compiles characteristic and measured properties of flows and sends flow reports to internal or external flow collectors. IPFIX is configurable via the command-line or REST interfaces. With IPFIX, customers configure flow records with match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports.

Compatibility with Traffic Insight

The AOS-CX traffic insight feature allows monitoring of large amount of data that it collects from various flow exporters like IPFIX, and provides the ability to filter, aggregate, and sort the data based on user flow monitor requests. Traffic insight tracks different monitor requests simultaneously and provides monitor reports per request. For more information on configuring the Traffic Insight features, refer to the *AOS-CX Security Guide*.

Compatibility with Application Recognition

If the **application recognition** feature is enabled, then the application data and the flow properties collected by AR and IPFIX are exported to external or internal IPFIX collectors. For more information on configuring the Application Recognition and Traffic Insight features, refer to the *AOS-CX Security Guide*.

Information Elements

The IPFIX Information Elements (IE) are entities that are defined and maintained by the Internet Assigned Numbers Authority (IANA). They are characterized by a unique piece of information they can provided about a flow. Information Elements may be either private or public. Private Information Elements are exported with a Private Enterprise Number (PEN).

AOS-CX can act as an intermediate collecting process for flow reports from hardware to append certain additional IPFIX information elements to the flow reports. When configured, the software will act as an intermediate exporting process to export the augmented flow reports to any configured flow exporters.

Standard Information Elements

octetDeltaCount
packetDeltaCount
protocolIdentifier
sourceTransportPort
sourceIPv4Address

Private Information Elements

(No private information elements are currently supported by the 5420 Switch series)
tlsClientVersion (ID 1029, 4823 Aruba)
tlsServerVersion (ID 1030, 4823 Aruba)
ja3 (ID 1031, 4823 Aruba)
ja3s (ID 1032, 4823 Aruba)

Standard Information Elements

ingressInterface
destinationTransportPort
destinationIPv4Address
sourceIPv6Address
destinationIPv6Address
classId
sourceMacAddress
vlanId
ipVersion
destinationMacAddress
applicationDescription
applicationId
applicationName
flowEndReason
flowStartMicroseconds
flowEndMicroseconds
dnsResponseCode
ingressPhysicalInterface
egressPhysicalInterface
applicationCategoryName

Private Information Elements

supportedNextProtocol (ID 1033, 4823 Aruba)
CertificateIssuer (ID 1034, 4823 Aruba)
CertificateIssueDate (ID 1035, 4823 Aruba)
CertificateExpiryDate (ID 1036, 4823 Aruba)
applicationType (ID 1100, 4823 Aruba)

About individual Information Elements

The following IEs are used for both IPv4 and IPv6 flows:

- cngSourceIP
- cngDestinationIP

They are always 16 bytes wide. The IP version of the flow is indicated in the **ipVersion** IE. For IPv4 addresses, the first 12 octets are zero and the IPv4 address is encoded in the last 4 octets. For IPv6 addresses, the full 16 bytes encode the address.

The following IEs reflect the hardware interface ID used by the associated flow:

- ingressPhysicalInterface
- egressPhysicalInterface

The hardware interface ID does not match the front-panel port number of the interface. The hardware interface ID can be mapped to the front-panel port number by using the REST API to query for **hw_intf_info** for all interfaces and saving the **switch_intf_id** value together with the associated front-panel name to form a mapping table.

Example using curl and jq:

```
bash
$ curl -X GET
  "{SWITCH}/rest/{VERSION}/system/interfaces?attributes=hw_
  intf,info,name&depth=2"
  -H "x-csrf-token: {CSRF_TOKEN}"
  -b {AUTH_COOKIE_FILE}
  | jq 'to_entries
      | map({port: .key, switch_intf_id: (.value.hw_intf_info.switch_intf_id |
  tonumber)})
      | sort_by(.switch_intf_id)
      | map({(.port): .switch_intf_id})
      | add'
{
  "1/1/6": 1,
  "1/1/2": 2,
  "1/1/1": 3,
  ...
}
```

Drop reason

Example Cause

Flow monitors

A flow monitor is applied to an interface to perform network traffic monitoring. A flow monitor consists of a flow record, a flow cache, and optional flow exporters. A flow record must be created and assigned to the flow monitor for the monitoring process to function. Flow data is compiled from the network traffic on the interface and stored in the flow cache based on the match (key) and collect (non-key) fields in the flow record. Data from the flow cache is exported by the flow exporters assigned to the flow monitor.

Flow Records

A flow record on 5420, 8100, 8360, 8325, 8325P, and 9300S Switch series defines match (key) fields and collection (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collection fields are the set of fields that identify information to collect for a flow, such as packet and byte counters. On the 5420, 6200, 6300, 6400, 8100, and 8360 switch series a maximum of sixteen flow records can be created.

There are six mandatory match fields, of which the IP match fields must be of the same type (IPv4 or IPv6).



A flow record is invalid if it does not contain one of the supported sets of match fields.

The supported sets of match fields on 5420, 8100 and 8360 Switch series are:

1. IPv4:

- IPv4 version
- IPV4 source address
- IPv4 destination address
- IPv4 protocol
- Transport destination port
- Transport source port

2. IPv6:

- IPv6 version
- IPv6 source address
- IPv6 destination address
- IPv6 protocol
- Transport destination port
- Transport source port

Flow Exporters

A flow exporter defines where and how to export flow reports. Flow exporters are created as standalone entities in the **config** context to provide flow monitors the ability to export flow reports.

Destinations

The destination specifies where flow reports are sent. There is one possible destination for a flow exporter within the 5420 and 6200 switch series:

1. Hostname or IP address of a device with an optional VRF

A flow exporter can only send flow reports to one destination. The destination type specifies which destination to use. If no destination type is specified, the default destination type is the hostname or IP address of a device with an optional VRF. If a VRF is not specified, the default VRF will be used. A destination of each type can be configured, but only the one corresponding to the destination type is used. If there is no destination corresponding to the destination type, then the flow exporter configuration is incomplete. If a new destination of a particular type is configured, it will replace the destination of that type that was previously configured.

Configuring IP Flow Information Export on 5420 and 6200 Switches

The following list describes the steps required to configure a IP flow information export (IPFIX) solution:

- Step one: Create flow records
- Step two: Configure flow exporter(s)
- Step three: Configure monitor(s)
- Step four: Apply a flow monitors to interface(s)



IPv6 related commands are only applicable to switches that support IPv6 protocol.

Step one: Create Flow Records

Flow Records are used to define the data that will be added to the IPFIX template. This example configures one record for IPv4 and one for IPv6.

```
switch(config)# flow record flowRecordv4
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
switch(config-flow-record)# collect application https url

switch(config)# flow record flowRecordv6
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 version
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# match transport source port
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect timestamp absolute first
```

Next, use the **show flow record** command to verify the configuration.

```
-----
Flow record 'flowRecordv4'
-----
Description           : ipv4
Status                : Accepted
Match Fields
  ipv4 destination address
  ipv4 protocol
  ipv4 source address
  ipv4 version
  transport destination port
  transport source port
Collect Fields
  application name
  counter bytes
  counter packets
  application https url
  timestamp absolute first
  timestamp absolute last
-----

Flow record 'flowRecordv6'
```

```

-----
Description          : ipv6
Status               : Accepted
Match Fields
  ipv6 destination address
  ipv6 protocol
  ipv6 source address
  ipv6 version
  transport destination port
  transport source port
Collect Fields
  application name
  counter bytes
  counter packets
  application https url
  timestamp absolute first
  timestamp absolute last

```

Step two: Configure flow exporter(s)

In this step, you can define an exporter to send to an external destination by hostname or IP address, or to an internal destination such as Traffic Insight. The example below configures IPFIX to export data to an external address/hostname:

```

switch(config)# flow exporter flowExternal
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 11.1.1.1
switch(config-flow-exporter)# show flow exporter
-----
Flow exporter 'flowExternal'
-----
Status          : Accepted
Export Protocol  : ipfix
Destination Type : Hostname or IP address
Destination      : 11.1.1.1
Transport Configuration
Protocol         : udp
Port            : 4739

```

To configure IPFIX to export to Traffic Insight, first configure Traffic Insight.

```

switch(config)# traffic-insight TI
switch(config-ti-TI)# source ipfix
switch(config-ti-TI)# monitor topN type topN-flows
switch(config-ti-TI)# monitor appFlow type application-flows
switch(config-ti-TI)# enable

```

Next, configure the flow exporter for Traffic Insight

```

switch(config)# flow exporter flowExpTI
switch(config-flow-exporter)# export-protocol ipfix
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight TI

```

You can use the **show flow exporter** command to verify the flow exporter configuration for Traffic Insight

```
switch(config)# show flow exporter flowExpTI
```

```
-----  
Flow exporter 'flowExpTI'  
-----
```

```
Status                : Accepted  
Export Protocol        : ipfix  
Destination Type       : Traffic Insight  
Destination            : TI  
Transport Configuration  
Protocol               : udp  
Port                   : 4739
```

Finally, use the **show run traffic-insight** command to verify the Traffic Insight configuration:

```
switch(config)# show running-config traffic-insight  
traffic-insight TI  
enable  
source ipfix  
!  
monitor topN type topN-flows entries 5  
monitor appFlow type application-flows
```

Step three: Configure the monitor(s)

First, configure an IPv4 flow monitor.

```
switch(config)# flow monitor flowMonv4  
switch(config-flow-monitor)# record flowRecordv4  
Switch (config-flow-monitor)# exporter flowExternal  
switch(config-flow-monitor)# exit
```

Next, configure an IPv6 flow monitor.

```
switch(config)# flow monitor flowMonv6  
switch(config-flow-monitor)# record flowRecordv6  
switch(config-flow-monitor)# exporter flowExternal  
switch(config-flow-monitor)# exit
```

Once both flow monitors are created, use the **show flow monitor** command to verify the flow monitor configurations.

```
switch(config-flow-monitor)# show flow monitor
```

```
-----  
Flow monitor 'flowMonv4'  
-----
```

```
Status                : Accepted  
Flow Record           : flowRecordv4  
Flow Exporter(s)       : flowExternal  
Cache Configuration  
Inactive Timeout      : 30  
Active Timeout         : 1800
```

```
-----  
Flow monitor 'flowMonv6'
```

```
-----  
Status                : Accepted  
Flow Record           : flowRecordv6  
Flow Exporter(s)      : flowExternal  
Cache Configuration  
Inactive Timeout      : 30  
Active Timeout        : 1800
```

A collecting process is not configured with IPFIX.

Role-based IPFIX

If IPFIX monitoring is configured on a port, a network administrator who wants to monitor traffic from clients must preconfigure IPFIX monitor before onboarding the clients. Also, if a client moves from one port to another, the monitor must be reconfigured accordingly. To reduce all these complexities, 5420, 6200, 6300, and 6400 switch series allow you to configure an IPFIX monitor on a port access role.

An IPFIX deployment with monitoring configured on a port uses TCAM to get flow statistics. Since TCAM is a high-demand resource which is shared across multiple protocols including the security protocols, security policy applications could fail if IPFIX consumes most of the TCAM resources. Port access clients can use a large amount of TCAM resources for policy applications, making it difficult support port access clients with TCAM based IPFIX.

With role-based IPFIX, when a user plugs a device into to a specific port, their user role is applied to the port. Whenever the user plugs that device into to a different port again, the role is applied to that specific port. This gives mobility and accessibility to the user and provides permissions specific to the user's role when the user's device moves across ports. Role-based IPFIX enables colorless port behavior.

There are some mutually exclusive features that will disable role-based IPFIX when they are enabled on the 6200, 6300, and 6400 platforms. The mutually exclusive features are:

- UBT
- MAC Lockout
- Extended Router MAC
- Source Port Filter
- IP Lockdown
- L3VNI

The following behaviors are observed in a deployment with role-based IPFIX:

- If a user applies the IPFIX monitor configuration on the port, it uses the existing TCAM infrastructure.
- If a port has both Role based IPFIX and Port based IPFIX, the Port based IPFIX implementation takes the priority.
- If IPFIX monitor configuration on a port moves from role based to TCAM based or vice-versa, existing IPFIX flows learned on that port are deleted.
- When the user role has an IPFIX configuration and fails to apply it, port access logs it, but it does not prevent the client from getting authorized.
- If port access is in the client mode and if there is more than one client with a conflicting IPFIX monitor configuration, traffic from all the clients will be exported through the IPFIX monitor configured for the first client.
- Role-based IPFIX and TCAM port-based IPFIX implementations can co-exist on a switch.
- Flows from unauthorized clients are not monitored.

These are some limitations to role-based IPFIX:

- Role based IPFIX can be enabled only through port access role assignment.
- Only Ingress flow monitoring is supported on both role-based and port-based IPFIX.

FAQs and Troubleshooting

- The following messages are displayed to indicate an illegal argument:
 - % The flow exporter <EXPORTER-NAME> does not exist.
 - % The flow record <RECORD-NAME> does not exist.
 - % The flow monitor <MONITOR-NAME> does not exist.
 - Invalid destination IP address or hostname entered.
 - Unable to create the flow exporter. The maximum allowed number of flow exporters (<max>) has been reached.
 - Unable to create the flow record. The maximum allowed number of flow records (<max>) has been reached.
 - Unable to create the flow monitor. The maximum allowed number of flow monitors (<max>) has been reached.
 - Flow monitor cannot be applied while interface is part of LAG <LAG-NAME>.
 - Flow monitor could not be applied.
 - Flow monitor could not be unapplied.

diag-dump ipfix basic

diag-dump ipfix basic

Description

Displays diagnostic information for IPFIX.

Examples

```
diag-dump ipfix basic
=====
[Start] Feature ipfix Time : Tue Apr 11 02:23:03 2023
=====
-----
[Start] Daemon ipfixd
-----
- IPFIX Record Cache dump -
- IPFIX Record ipfix -

....

:- IPFIX Monitor v6ti completed -
- End of IPFIX Monitor Cache dump -
-----
[End] Daemon ipfixd
-----
-----
[Start] Daemon ops-switchd
-----
Key format: <traffic_type>_<coalescence_id>_<agent_id>_<asic_port>
```

```

Key                                     TCAM Entry ID   Count
-----
1_1532781829_3_20                     0xffff7c7e7a00  1
1_3217499901_1_12                     0xffff91187580  1
1_3217499901_1_13                     0xffff91183d80  1
1_3217499901_1_14                     0xffff91186e80  1
....

-----
[End] Daemon ops-switchd
-----
=====
[End] Feature ipfix
=====
Diagnostic-dump captured for feature ipfix

```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

description

```

description <DESCRIPTION>
no description <DESCRIPTION>

```

Description

Configures the description for the following telemetry options:

- flow monitor in the **config-flow-monitor** context
- flow congestion monitor in the **config-flow-congestion-monitor** context
- flow exporter in the **config-flow-exporter** context
- flow record in the **config-flow-record** context

The no form deletes a flow congestion monitor.

Parameter	Description
<DESCRIPTION>	Displays a string of 256 characters, maximum, including spaces.

Examples

Adding or modifying the description of flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1  
switch(config-flow-monitor)# description Used for analyzing basic ipv4 traffic
```

Adding or modifying the description of flow record, **flow-record-1**:

```
switch(config)# flow record flow-record-1  
switch(config-flow-record)# description Used for basic traffic analysis
```

Adding or modifying the description of flow exporter, **flow-exporter-1**:

```
switch(config)# flow exporter flow-exporter-1  
  
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
```

Removing the description of flow monitor, **flow-monitor-1**:

```
switch(config)# flow exporter flow-monitor-1  
switch(config-flow-exporter)# no description
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
5420	config-flow-record config-flow-monitor config-flow-congestion-monitor config-flow-exporter	Administrators or local user group members with execution rights for this command.

exporter

exporter <EXPORTER-NAME>
no exporter <EXPORTER-NAME>

Description

Assigns a flow exporter to **flow monitor** in the **config-flow-monitor** context and **flow congestion monitor** in the **config-flow-congestion-monitor** context.
The no form removes the flow exporter from the selected monitor.

Parameter	Description
<EXPORTER-NAME>	Specifies the name of the flow exporter assigned to the monitor.



A maximum of one flow exporter can be applied to each flow monitor, and one flow exporter can be applied to each flow congestion monitor.

Examples

Assigning flow exporter, **flow-exporter-1**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow monitor, **flow-monitor-1**:

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-monitor)#
```

Assigning flow exporter, **flow-exporter-1**, to flow congestion monitor, **congestion-monitor-1**:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# exporter flow-exporter-1
```

Removing a flow exporter assigned to flow congestion monitor:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# no exporter flow-exporter-1
```

Attempting to assign non-existent flow exporter, **flow-exporter-5**, to flow congestion monitor, **congestion-monitor-1**:

```
switch(config)# flow congestion-monitor congestion-monitor-1
switch(config-flow-congestion-monitor)# exporter flow-exporter-5
Flow exporter 'flow-exporter-5' does not exist.
switch(config-flow-congestion-monitor)#
```

Attempting to assign flow exporter, **flow-exporter-6**, to monitor, **flow-monitor-3**, when two exporters are assigned:


```

switch(config)# flow monitor flow-monitor-3
switch(config-flow-monitor)# exporter flow-exporter 4
switch(config-flow-monitor)# exporter flow-exporter 5
switch(config-flow-monitor)# exporter flow-exporter 6
Cannot assign more than two flow exporters to a flow monitor.
switch(config-flow-monitor)#

```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
5420	config-flow-monitor config-flow-congestion-monitor	Administrators or local user group members with execution rights for this command.

flow exporter

```

flow exporter <name>
  description <description>
  destination
    <hostname>
    [vrf <vrfname>]
  <ipaddr>
    [vrf <vrfname>]
  <ip6addr>
    [vrf <vrfname>]
  type
    hostname-or-ip-addr
    traffic-insight}
  traffic-insight <instance-name>
no ...
template data timeout <timeout>
transport udp <port>

```

Description

A flow exporter is the part of the IP Flow Information Export (IPFIX) feature that defines how a flow monitor exports flow reports. You can assign the same flow exporter configuration to more than one flow monitor. Each flow exporter includes a destination setting that identifies the device to which the flow reports are sent

Parameter	Description
<name>	Name of the flow exporter, up to 64 characters.

Parameter	Description
<code>description <description></code>	A description of the flow exporter, up to 256 characters and spaces.
<code>destination</code>	Configure the export destination
<code><hostname></code>	The exporter sends flow records to the specified hostname destination. The hostname can be a string of up to 64 characters.
<code><IPaddr></code>	The exporter sends flow records to this IPv4 address destination.
<code><ip6addr></code>	The exporter sends flow records to this IPv6 address destination.
<code>type</code>	Configure the type of the destination.
<code>hostname-or-ip-addr</code>	Define the destination type as a hostname or IP address.
<code>traffic-insight <name></code>	Define the destination type as a traffic insight instance.
<code>no ...</code>	Negate any configured parameter.
<code>traffic-insight <INSTANCE-NAME></code>	Specify the a Traffic Insight instance to be used as the destination.
<code>template data timeout <timeout></code>	A flow exporter template describes the format of exported flow reports. Therefore, flow reports cannot be decoded properly without the corresponding templates. This setting defines how often the flow exporter will resend templates to the flow monitor. The supported range is 1-86400 seconds, and the default is 600 seconds.
<code>transport udp <port></code>	Transport protocol and port for sending flow record reports. The default port is port 4739.

Usage

The following table shows the maximum supported of flow monitors and flow exporters for each switch model.

Switch	Maximum Flow Monitors	Maximum Flow Exporters
5420	16	Two flow exporters can be applied to a single flow monitor.

Examples

The following example creates a flow exporter configuration named **exporter-1**.

The following example sets a Traffic Insight instance as the destination for a flow exporter

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight instance-1
```

The following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use:

```
switch(config)# flow exporter exporter-4
switch(config-flow-exporter)# destination collector-1
switch(config-flow-exporter)# destination traffic-insight instance-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

The following example sets an IPv4 address as the destination for a flow exporter

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 192.168.0.1
```

The following example sets a hostname as the destination for a flow exporter

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination collector1
```

The following example sets an IPv6 address as the destination for a flow exporter

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 2001:db87::8a2e:370a:7334
```

The following example sets an IPv4 address as the destination for a flow exporter with the VRF to which the IPv4 address belongs

```
switch(config)# flow exporter exporter-2
switch(config-flow-exporter)# destination type hostname-or-ip-addr
switch(config-flow-exporter)# destination 192.0.2.1 vrf VRF1
```

The following example sets a Traffic Insight instance as the destination for a flow exporter

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# destination type traffic-insight
switch(config-flow-exporter)# destination traffic-insight instance-1
```

The following example adds a destination of each possible type and set **hostname-or-ip-addr** as the type to use

```
switch(config)# flow exporter exporter-4
switch(config-flow-exporter)# destination collector-1
switch(config-flow-exporter)# destination traffic-insight instance-1
switch(config-flow-exporter)# destination type hostname-or-ip-addr
```

The following example removes the destination of type **traffic-insight** from a flow exporter:

```
switch(config)# flow exporter exporter-3
switch(config-flow-exporter)# no destination traffic-insight
```

The following example removes the destination of type **hostname-or-ip-addr** from a flow exporter

```
switch(config)# flow exporter exporter-1
switch(config-flow-exporter)# no destination
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config config-flow-exporter	Administrators or local user group members with execution rights for this command.

flow monitor

```
flow monitor <name>
  exporter <name>
  cache timeout {inactive <timeout>} [{active <timeout>}]
  description <description>
  record <name>
```

Description

On HPE Aruba Networking 5420, 6200, 6300, 6400, 8325, 8325H, 8325P, 8100, 8360, 9300, 9300S Switch series, a flow monitor is the part of the IP Flow Information Export (IPFIX) feature that performs network monitoring for the selected interface. A flow monitor configuration consists of a flow record, a flow cache, and one or more associated flow exporters. A flow monitor compiles data from the network traffic on the interface and stores it in the flow cache in a format defined by the flow record. The flow exporters associated with the monitor then export data from the flow cache to the flow exporter destination.



HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8360 Switch series support a maximum of sixteen flow monitors with a limit of two flow exporters that can be applied to a single flow monitor. If no software augmentation of flows is required, there is no need to configure a flow collector or flow monitor.

Parameter	Description
<name>	Name of the flow monitor, up to 64 characters.
cache timeout active <timeout>	Use the cache timeout parameter to define an active or inactive timeout for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout. The active timeout range is 30-604800. The default active time out value is 1800 and inactive timeout value is 30.
cache timeout inactive <timeout>	Use the cache timeout parameter to define an inactive timeout

Parameter	Description
	for the flow monitor. A flow monitor closes a flow session that is active for longer than the active timeout or inactive for longer than the inactive timeout. For 5420, 6200, 6300, 6400, 8100, 8360, 9300S Switch Series, the inactive timeout range is 30-604800. The default active time out value is 1800 and inactive timeout value is 30.
description	A description up to 256 characters long, including spaces.
exporter <name>	Assign a flow exporter to a flow monitor.
record <name>	(For HPE Aruba Networking 5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 9300, 9300S Switch series) Assigns a flow record to a flow monitor.

Examples

The following example creates a flow monitor configuration named **monitor-1**.

```
switch(config)# flow monitor monitor-1
switch(config-flow-monitor)# description Monitor for analyzing basic ipv4 traffic
switch(config-flow-monitor)# exporter flow-exporter-1
switch(config-flow-monitor)# record flow-record-1
switch(config-flow-monitor)# cache timeout inactive 120
switch(config-flow-monitor)# cache timeout active 1500
```

The following workflow changes the flow record assigned to a flow monitor.

```
switch(config)# flow monitor flow-monitor-1
switch(config-flow-monitor)# record flow-record-2
```

Create more than the maximum number of allowed flow monitors

```
switch(config)# flow monitor monitor-1
switch(config)# flow monitor monitor-2
<--OUTPUT OMITTED FOR BREVITY-->
switch(config)# flow monitor monitor-16
switch(config)# flow monitor monitor-17
No more than 16 flow monitors can be configured. Another flow monitor
must be removed first.
```

Assign more than one flow exporter to flow monitor ****flow-monitor-2***

```
switch(config)# flow monitor flow-monitor-2
switch(config-flow-monitor)# exporter flow-exporter-2
switch(config-flow-monitor)# exporter flow-exporter-3
```

Add or modify the description of flow record ****flow-record-1****

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# description Used for basic traffic analysis
```

Add or modify the description of flow exporter ****flow-exporter-1****

```
switch(config)# flow exporter flow-exporter-1
switch(config-flow-exporter)# description Exports flows to 10.2.3.45:2055
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

flow record

```
flow record <record-name>
  match
    ip {source|destination} address
    ip protocol|version
    ipv6 {source|destination} address
    ipv6 protocol|version
    transport {source | destination} port
  collect
    application name
    application https url
    application dns response-code
    forwarding-status
    egress vlan
    egress interface
    egress queue
    counter {packets|bytes}
    transport {source | destination} port
    timestamp absolute first
    timestamp absolute last
  description <description>
```

Description

Creates or modifies a flow record and switches to the **config-flow-record** context for the flow record. Define data to be included in a flow record by configuring flow record match and collect fields.

A flow record defines match (key) fields and collection (non-key) fields. Customers configure flow records with **match** (key) fields and **collect** (non-key) fields. Match fields are the set of fields that define a flow, such as IP address or UDP port. Collect fields are the set of fields that identify information to collect for a flow, such as packet and byte counters.

Traffic with matching attributes (for example, traffic coming from the same interface, sent to the same destination with the same protocol) are classified as a single flow. Information for some or all of the matched settings can be collected and exported to a destination defined by the flow exporter assigned to the flow monitor.



Traffic must match a match rule definition before it can be collected and sent. You cannot collect and send data that is not matched.



A maximum of one flow record can be created for 5420, 8325, 8325H, 8325P, 930 and 9300S Switch series. For 6200, 8360, 8100, 6300, and 6400 Switch series, a maximum of 16 flow records can be created.

Parameter	Description
<record-name>	Name of the flow monitor, up to 64 characters.
match	match traffic according to one or more of the following key attributes: ▪ ip source: Match traffic from the same IPv4 source. ▪ ip destination: Match traffic to the same IPv4 destination. ▪ ip protocol: Match traffic using the same IP version ▪ ip version: Match traffic using the same IP protocol ▪ ipv6 source: Match traffic from an IPv6 source. ▪ ipv6 destination: Match traffic to an IPv6 destination. ▪ ipv6 protocol: Match traffic using the same IPv6 version ▪ ipv6 version: Match traffic using the same IPv6 protocol ▪ transport {source destination} port: Match traffic by source or destination transport port
description	A description for the flow record up to 256 characters long, including spaces.
collect	Configures data fields to be included a flow record. ▪ application name: Specify the application name as a non-key field in a flow record. ▪ application https url: Specify the HTTP/HTTPS application URL as a non-key field in a flow record. ▪ application dns response-code: Specify the DNS parameters and DNS response code as a non-key field in the flow record. ▪ application tcp establishment-time: Specifies TCP connection establishment time as a non-key field in a flow record ▪ counter bytes: Collect counter data for bytes in the flow. ▪ counter packets: Collect counter data for packets in the flow. ▪ egress interface: Specifies an egress interface as a non-key field in a flow record. ▪ egress queue: Specifies an egress queue as a non-key field in a flow record. ▪ egress vlan: Specifies an egress VLAN ID as a non-key field in a flow record. ▪ fowarding status: Specifies forwarding status as a non-key field in a flow record ▪ timestamp absolute first: Collect absolute timestamp of the first packet observed. ▪ timestamp absolute last: Collect absolute timestamp of the last packet observed.

Examples

```
switch(config-flow-record)# match ip version
```

Adding IPv4 match fields to flow record ****flow-record-1**** using the ``ip`` keyword

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
```

Adding IPv6 match fields to flow record ****flow-record-2****

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# match ipv6 source address
switch(config-flow-record)# match ipv6 destination address
switch(config-flow-record)# match ipv6 protocol
switch(config-flow-record)# match ipv6 version
```

Adding IPv6 collect fields to flow record ****flow-record-2****

```
switch(config)# flow record flow-record-2
switch(config-flow-record)# collect application tcp establishment-time
switch(config-flow-record)# collect egress-vlan
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect forwarding-status
switch(config-flow-record)# collect egress queue
```

Adding IPv4 match fields to flow record **flow-record1**

```
switch(config)# flow record flowrecord1
switch(config-flow-record)# match ip source address
switch(config-flow-record)# match ip destination address
switch(config-flow-record)# match ip protocol
switch(config-flow-record)# match ip version
switch(config-flow-record)# match transport source port
switch(config-flow-record)# match transport destination port
switch(config-flow-record)# collect application name
switch(config-flow-record)# collect counter bytes
switch(config-flow-record)# collect counter packets
switch(config-flow-record)# collect egress interface
switch(config-flow-record)# collect egress queue
switch(config-flow-record)# collect egress vlan
switch(config-flow-record)# collect forwarding-status
switch(config-flow-record)# collect application https url
switch(config-flow-record)# collect application tcp establishment-time
switch(config-flow-record)# collect timestamp absolute first
switch(config-flow-record)# collect timestamp absolute last
```

Removing the IPv4 destination address match field from flow record ****flow-record-1**** using the ``ip`` keyword

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ip destination address
```


Removing the IPv4 destination address match field from flow record **flow-record-1** using the `ipv4` keyword

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match ipv4 destination address
```

Removing the transport destination port match field from flow record **flow-record-1**

```
switch(config)# flow record flow-record-1
switch(config-flow-record)# no match transport destination port
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config config-flow-record	Administrators or local user group members with execution rights for this command.

flow-tracking

```
flow-tracking enable
no flow tracking
  icmp-ageout
  interface-flow-limit
  flow statistics enable
  tcp-ageout
  track icmp
  udp-ageout
```

Description

Configures flow tracking for TCP and UDP flows, and optionally, ICMP flows. The **no** form of this command deletes the flow tracking configuration context.

In order to optimize the flow removal process, flows that have aged-out are flushed in batches. A flow that has aged out is flushed only when the next batch processes. This can cause some flows to stay inactive for a slightly longer time than the value configured here.

Parameter	Description
enable	Enables flow tracking.
icmp-ageout	Configures an age-out time for ICMP flows, in seconds. Range: 10-86400. Default: 15.
interface-flow-limit	Configures global concurrent flow limit for flow tracking enabled

Parameter	Description
	interfaces. Range: For 5420 Switch series,64-10000. Default: none.
flow statistics	Enable flow statistics collection.
tcp-ageout	Configures age-out time for established TCP flows in seconds. Range: 120-86400. Default: 600.
track icmp	Enable tracking of ICMP flows, in addition to the TCP/UDP flows tracked by default.
udp-ageout	Configures age-out time for established UDP flows in seconds. Range: 30-86400. Default: 30.

Examples

Configuring flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)#
```

Deleting flow tracking:

```
switch(config)# no flow-tracking
switch(config)#
```

Enabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# enable
```

Disabling flow tracking:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no enable
```

Configuring an established ICMP flow age-out to 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# icmp-ageout 600
```

Removing an established ICMP flow age-out of 600 seconds:

```
switch(config)# flow-tracking
switch(config-flow-tracking)# no icmp-ageout 600
```

Configuring an established TCP flow age-out to 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # tcp-ageout 1000
```

Removing an established TCP flow age-out of 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no tcp-ageout 1000
```

Configuring an established UDP flow age-out to 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # udp-ageout 1000
```

Removing an established UDP flow age-out of 1000 seconds:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no udp-ageout 1000
```

Configuring global level interface flow limit to 256 interfaces:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # interface-flow-limit 256
```

Removing global level interface flow limit to 256 interfaces:

```
switch(config) # flow-tracking  
switch(config-flow-tracking) # no interface-flow-limit 256
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

ip|ipv6 flow monitor (interface)

[no] ip|ipv6 flow monitor (interface)

Description

Enable flow monitoring on inbound and outbound interfaces by assigning a flow monitor to that interface. Only physical interfaces and LAG interfaces can be monitored. A flow monitor cannot be applied to an interface that is part of a LAG. If an unsupported application is attempted, an error

message will be displayed. If the flow monitor is associated with a flow record that contains application fields as collect fields, then Application Recognition should be enabled on the same interface.

The **[no]** form of command disables the flow monitoring.

Examples

Enable a flow monitor configuration named **flow-monitor-1** for IPv4 traffic on a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip flow monitor flow-monitor-1 in
```

Associate a flow monitor configuration named **flow-monitor-2** for IPv4 traffic on a LAG interface.

```
switch(config)# interface lag 1
switch(config-if)# ip flow monitor flow-monitor-2 in
```

Associate a flow monitor configuration named **flow-monitor-3** for IPv6 traffic on a physical interface.

```
switch(config)# interface 1/1/1
switch(config-if)# ip flow monitor flow-monitor-3 in
```

Associate a flow monitor configuration named **flow-monitor-4** for IPv6 traffic on a physical interface.

```
switch(config)# interface lag 1
switch(config-if)# ipv6 flow monitor flow-monitor-1 in
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

ipv4|ipv6 flow monitor (role)

[no] ip|ipv6 flow monitor <name>

Description

Enable flow monitoring on a role. The authorization status of a client does not depend on the flow monitor status for the client in hardware. The client will be authorized even if the system is not able to apply the flow monitor configuration in the role.

In case of multiple clients onboard to a port with varied flow monitor configurations, the flow monitor associated with the first authenticated client on the port will be applied for all the traffic on the port.

The **[no]** form of command removes the flow monitor from the role.

Parameter	Description
<name>	Name of the flow monitor , up to 64 characters.

Examples

Enable an IPv4 flow monitor on a role named role01

```
switch# config terminal
switch(config)# port-access role role01
switch(config-pa-role)# ip flow monitor flow-monitor-1
```

Enable an IPv6 flow monitor on a role named role01

```
switch# config terminal
switch(config)# port-access role role01
switch(config-pa-role)# ipv6 flow monitor flow-monitor-2
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config config-pa-role	Administrators or local user group members with execution rights for this command.

show flow exporter

```
show flow exporter [<name>] [statistics] [vsx-peer]
```

Description

Displays flow exporter configuration and status. When no exporter name is specified, the output of this command displays information for all flow exporters.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: exporter does not exist)
- Rejected (Internal error: destination type does not exist)
- Rejected (Destination type is hostname or IP address, but no destination is specified)
- Rejected (Destination type is hostname or IP address, but the specified hostname or IP address is invalid)
- Rejected (Destination type is Traffic Insight, but no destination is specified)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance does not exist)
- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance is not enabled)

- Rejected (Destination type is Traffic Insight, but the specified Traffic Insight instance source is not IPFIX)
- Rejected (Internal error: destination type is Traffic Insight, but the specified Traffic Insight instance is invalid)
- Rejected (Internal error: the specified destination VRF is invalid)
- Rejected (Internal error: the specified destination VRF is not ready)

Parameter	Description
<name>	Name of the flow exporter.
statistics	Adds statistical information about the flow exporter to the output.
vsx-peer	Displays flow collector configuration for the VSX peer.

Examples

Display the configuration of a flow exporter named **exporter-1**.

```
switch# show flow exporter exporter-1
-----
Flow exporter 'exporter-1'
-----
Description           : Exports to the first collector
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Hostname or IP address
Destination            : 192.168.0.1
Transport Configuration
  Protocol              : UDP
  Port                  : 9995
```

Display statistics information for all flow exporters

```
switch# show flow exporter statistics
-----
Flow exporter 'exporter-1'
-----
Reports sent           : 14961
-----
Flow exporter 'exporter-2'
-----
Reports sent           : 5
```

Display the configuration of all flow exporters:

```
switch# show flow exporter
-----
Flow exporter 'exporter-1'
-----
Reports sent           : 0
```

Display information with no flow exporters configured

```
switch# show flow exporter
No flow exporters configured.
switch# show flow exporter statistics
No flow exporters configured.
```

Display a flow exporter's information with TI as a destination

```
switch# show flow exporter exporter-5
-----
Exporter Name           : exporter-5
-----
Description             : Exporter configured with TI as the destination
Status                  : Rejected (Destination type is Traffic Insight, but the
specified Traffic Insight instance does not exist)
Export Protocol         : ipfix
Destination Type        : Traffic Insight
Destination              : instance-1
Transport Configuration
Protocol                : UDP
Port                    : 2055
```

Display information for a flow exporter that has multiple destinations of different types configured and *hostname-or-ip-addr* is specified as the destination type to use

```
switch# show flow exporter exporter-6
-----
Flow exporter 'exporter-6'
-----
Description             : TI and hostname configured, but type is hostname-or-ip-
addr
Status                  : Accepted
Export Protocol         : ipfix
Destination Type        : Hostname or IP address
Destination              : collector-1
Destination VRF         : mgmt
Transport Configuration
Protocol                : UDP
Port                    : 4821
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

show flow monitor

```
show flow monitor [statistics]
show flow monitor <MONITOR-NAME>[statistics]
```

Description

Displays flow monitor configuration and status. When no monitor name is specified, the output of this command displays information for all flow monitors.

Parameter	Description
<name>	Name of the flow monitor.
statistics	For 5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 9300, 9300S Switch series, include the statistics parameter to display additional flow and cache statistics.

Usage

The output of this command can indicate the following status types:

- Possible status types for all switches
 - Accepted
 - Rejected (Internal error: monitor does not exist)
 - Rejected (The state of one or more of the assigned flow exporters is rejected)
- Possible Status types for 5420, 6200, 6300, 6400, 8100, 8325, 8325P, 8325H, 8360, 9300, 9300S Switch series:
 - Rejected (A record must be assigned to the monitor, but no record is assigned)
 - Rejected (The state of the assigned record is rejected)
 - Rejected (Internal error: failure in processing the record configuration)

		Switches that support this statistic
Current Entries	Current number of flows in the flow cache for this flow monitor	5420
Flows Added	Total number of flows added to the flow cache for this flow monitor since it was created	5420
Total Flows Terminated	Total number of flows removed from the flow cache for this flow monitor since it was created due to any flow end reason	5420
Flows Aged	Number of flows removed from the flow cache for this flow monitor since it was created due to active or inactive timeout	5420
Inactive Timeout	Number of flows removed from the flow cache for this flow monitor since it was created due to inactive cache timeout.	5420

		Switches that support this statistic
Forced End	Number of flows removed from the flow cache for this flow monitor since it was created due to some external event. For example, the shutdown of a flow monitor.	5420

Examples

Display information for a flow monitor on a 5420, 9300, 9300S Switch series:

```
switch# show flow monitor monitor-1
-----
Flow monitor 'monitor-1'
-----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Flow Record          : record-1
Flow Exporter(s)     : exporter-1
Cache Configuration
Inactive Timeout     : 1800
Active Timeout       : 300
```

Display information and statistics for a flow monitor on 5420, 6200, 6300, 6400, 8100 or 8360 Switch series:

```
show flow monitor statistics
-----
Flow monitor 'monitor-1'
-----
Current Entries      : 2
Flows Added          : 6
Total Flows Terminated : 4
  Flows Aged         : 2
    Active Timeout   : 1
    Inactive Timeout : 1
  End of Flow Detected : 2
  Forced End         : 0
  Flows Aged         : 4
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	config config-flow-monitor	Administrators or local user group members with execution rights for this command.

show flow record

show flow record [<name>]

Description

Display flow record configuration and status. When no record name is specified, the output of this command displays information for all flow records.

The output of this command can indicate the following status types:

- Accepted
- Rejected (Internal error: failed to process record)
- Rejected (Mix of IPv4 and IPv6 match fields is not allowed. Specify match fields of the same IP version (IPv4 or IPv6))

Parameter	Description
<name>	Name of the flow record.

Examples



IPv6 related commands are only applicable to switches that support IPv6 protocol.

Display the configuration of a flow record named **flow-record-1**.

```
switch# show flow record record-1
-----
Flow record  'record-1'
-----
Description           : Used for IPv4 traffic analysis
Status                : Accepted
Match Fields
  ipv4 destination address
  ipv4 protocol
  ipv4 source address

  transport destination port
  transport source port
Collect Fields
  counter bytes
  counter packets
  application https URL
```

Display the information of a specific flow record.

```
switch# show flow record record-1
-----
Flow record  'record-1'
-----
Description           : Used for IPv4 traffic analysis
Status                : Accepted
Match Fields
  ipv4 destination address
  ipv4 protocol
```

```
    ipv4 source address

    transport destination port
    transport source port
Collect Fields
    application https url
    counter bytes
    counter packets
```

Display information for all flow records

```
switch# show flow record
-----
Flow record  'record-1'
-----
Description          : Used for IPv4 traffic analysis
Status               : Accepted
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
    transport destination port
    transport source port
Collect Fields
    counter bytes
    counter packets
-----
Flow record  'record-2'
-----
Description          : Used for IPv6 traffic analysis
Status               : Accepted
Match Fields
    ipv6 destination address
    ipv6 protocol
    ipv6 source address
    ipv6 version
    transport destination port
    transport source port
Collect Fields
    application name
    counter bytes
    counter packets
...
```

Display information for a specific flow record

```
switch# show flow record record-3
-----
Flow record  'record-3'
-----
Description          : Used for IPv4 traffic analysis
destination address, protocol, transport destination port, and transport source
port.)
Match Fields
    ipv4 destination address
    ipv4 protocol
    ipv4 source address
Collect Fields
    counter bytes
```

```
counter packets
```

Display information with no flow records configured

```
switch# show flow record
No flow records configured
```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

show flow-tracking

```
show flow-tracking
```

Description

Displays flow-tracking and statistics collection configurations and status.

Examples

Display the configuration of role based flow tracking on a 5420 or 6200 Switch series.

```
switch (config)# show flow-tracking
Flow Tracking Global Configuration
Configuration status      : Enabled
Operational status       : Enabled
Failure Reason           : NA
UDP Ageout               : 60  (Seconds)
TCP Ageout               : 600 (Seconds)
ICMP Ageout              : 15  (Seconds)
Interface Flow limit     : None
Tracked Protocols        : TCP, UDP
Statistics Collection
Configuration Status      : Enabled
Operational Status       : Enabled
Failure Reason           : NA
Flow Tracking Port Configuration
Interface      App Recognition      IPFIX      Operation Status
-----
1/1/1         Disabled                  Disabled   Disabled
1/1/2         Disabled                  Disabled   Disabled
1/1/3         Disabled                  Disabled   Disabled
1/1/4         Disabled                  Disabled   Disabled
```

1/1/5	Disabled	Disabled	Disabled
1/1/6	Disabled	Disabled	Disabled
1/1/7	Disabled	Disabled	Disabled
1/1/8	Disabled	Disabled	Disabled
1/1/9	Disabled	Disabled	Disabled
1/1/10	Disabled	Disabled	Disabled
1/1/11	Disabled	Disabled	Disabled
1/1/12	Disabled	Disabled	Disabled
1/1/13	Disabled	Disabled	Disabled
1/1/14	Disabled	Disabled	Disabled
1/1/15	Enabled	Disabled	Enabled
1/1/16	Disabled	Disabled	Disabled
1/1/17	Disabled	Disabled	Disabled
1/1/18	Disabled	Disabled	Disabled
1/1/19	Disabled	Disabled	Disabled
1/1/20	Disabled	Disabled	Disabled
1/1/21	Disabled	Disabled	Disabled
1/1/22	Disabled	Disabled	Disabled
1/1/23	Disabled	Disabled	Disabled
1/1/24	Disabled	Disabled	Disabled
1/1/25	Disabled	Disabled	Disabled
1/1/26	Disabled	Disabled	Disabled
1/1/27	Disabled	Disabled	Disabled
1/1/28	Disabled	Disabled	Disabled

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

show running-config

```
show running-config
    flow [ | record | interface [lag]][vsx-peer]
```

Description

Shows all IPFIX flow configuration.



To show the VSX peer switch configuration, insert **vsx-peer** at the end.

Parameter	Description
flow	Displays IPFIX flow configuration

Parameter	Description
record	Displays flow record configuration.
interface	Displays interfaces with flow monitors applied.
lag	Displays LAG interfaces with flow monitors applied.
vsx-peer	Displays VSX peer switch information.

Examples

Showing **show running-config flow**:

```
switch# show running-config flow
flow exporter exporter-1
  description This is an exporter
  destination 40.0.0.2 vrf default
  transport udp 2055
flow record record-1
  description This is a record
  match ipv4 destination address
  match ipv4 protocol
  match ipv4 source address
  match ipv4 version
  match transport destination port
  match transport source port
  collect counter bytes
  collect counter packets
  collect timestamp absolute first
  collect timestamp absolute last
  collect timestamp absolute last
flow monitor monitor-1
  description This is a monitor
  cache timeout active 30
  exporter exporter-1
  record record-1
interface lag 1
  ip flow monitor monitor-1 in
interface 1/1/1
  ip flow monitor monitor-1 in
```

Showing current IPFIX flow monitor configurations:

```
switch# show running-config flow monitor
flow monitor monitor-1
  description This is a monitor
  exporter exporter-1
  record record-1
```

Showing current IPFIX flow congestion monitor configurations:

```
switch# show running-config flow congestion-monitor
flow congestion-monitor cng-mon-1
  description This is a monitor
  exporter exporter-1
  ingress-interface 1/1/10
```

```
ingress-interface 1/1/11
```

Showing operational status of the current configuration:

```
switch# show running-config
flow exporter exporter-1
  description This is an exporter
  destination 40.0.0.2
  transport udp 2055
flow congestion-monitor monitor-1
  exporter exporter-1
  ingress-interface 1/1/1
interface 1/1/1
  ip-all flow congestion-monitor m out queue 0-7
    ! disabled - Blocked by a higher precedence feature
interface 1/1/2
  ip-all flow congestion-monitor m out queue 0-7
    ! disabled - Blocked by a higher precedence feature
```

Command History

Release	Modification
10.16	Command introduced

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

show tech ipfix

```
show tech ipfix
```

Description

Shows the IPFIX configuration settings.

If applicable source IP address or source interface is configured for the IPFIX protocol, that configuration is used.

Examples

The example shows the IPFIX configuration settings.

```
switch#show tech ipfix
=====
Show Tech executed on Tue Apr 11 02:43:06 2023
=====
[Begin] Feature ipfix
=====
*****
Command : show flow exporter
```

```

*****
-----
Flow exporter 'ipfix'
-----
Status                : Accepted
Export Protocol        : ipfix
Destination Type       : Traffic Insight
Destination            : tl
Transport Configuration
Protocol               : udp
Port                   : 4739
-----
Flow exporter 'V6E1'
-----

....

=====
[End] Feature ipfix
=====

```

Command History

Release	Modification
10.16	Command introduced on the 5420 switch series.

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

Overview

AOS-CX provides multiple features that assist with monitoring queuing behavior such as Queue Monitoring and an extension of it – Congestion Event Detection.

Queue monitoring

The Queue Monitoring feature provides an ongoing collection of queue depth, queue statistics and storage of those statistics, in a time-series database. The time-series data is available for viewing on-switch and provides a history of what has occurred in each of the monitored queues. Queue depth data, accessible on the switch for reviewing the queue statistics history, helps with identifying network issues or behavior patterns.

Queue statistics history

Queue monitoring stores collected queue depth data in a time-series database within the switch. This data, accessible on the switch for reviewing the queue statistics history, helps with identifying network issues or behavior patterns.

Queue depth

Queue depth is the memory size required to buffer packets in a queue during congestion. The value changes depending on the arrival rate of the packets destined to egress a given queue, and the rate at which that queue is able to transmit packets.

Data retention limits

Data retention limits are enforced within the switch by reducing time precision of the stored data within specific windows of time. The *retention duration* for queue statistics data is derived from the frequency of the polling interval.

Table 1: *Queue monitoring data retention schedule*

Polling interval	Maximum statistic age
10 seconds	8 hours



The switch has limited storage for retaining information. For longer monitoring durations, use an external monitor with the queue monitoring feature to collect and store data.

Using collected data

Queue monitoring provides the ability to look into switch congestion-related performance issues and identify when problems have occurred. The methods available for reviewing performance issues are:

- Congestion events
- Per-interface congestion histograms
- Per-interface congestion-detailed view

As anomalies are identified, the interface queue that experienced the anomaly is the starting point for identifying the cause of the behavior. Examine traffic patterns for flows competing for network bandwidth to determine if corrective action is needed to address potential problems.

Congestion event detection

The Congestion Event Detection feature is an extension of Queue Monitoring that allows users to configure alert thresholds for queue depth, queue rate, and queue drops. Alerts can be viewed on-switch and can also be sent to the switch event log on a selective basis.

Congestion event configuration

The congestion event detection feature is configured with the threshold condition(s) under which an event occurrence starts, ends, and is measured. Each congestion event profile is comprised of zero or more congestion event entries.

Congestion event profile

A congestion event profile entry has a **Valid** status if all the required fields are specified with values that pass validation rules, otherwise, the congestion event profile entry status shows as **Invalid**. Each congestion event profile entry must specify the following information:

- type
- trigger threshold
- reset threshold
- units for provided values
- retention group
- action (optional)

Using congestion events

A congestion event represents a single occurrence of conditions on an interface queue where a configured threshold was violated. Configured thresholds are specified as part of a congestion event profile and are applied on a per-interface basis along with queue monitoring. When a congestion event occurs, relevant data is collected and stored as an *event occurrence* within the congestion event retention group for later analysis. The switch provides 3 retention groups for storing event occurrences, to be used however desired. One effective method is to treat them as "low", "medium", and "high" severity event occurrence storage. This method allows to quickly check the switch for specific severity events using the **show congestion-event retention-group <NUMBER>** command.



Once a congestion event retention group has reached the limit of 1000 stored events, the oldest completed event occurrence in the group is discarded in order to store a new event.

boot set-default

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted. Changes to this setting may impact other features, such as the job scheduler feature configured via the **reload at** or **reload after** commands.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
<code>primary</code>	Selects the primary operating system image for this reboot and sets the configured default operating system image to primary for future reboots.
<code>secondary</code>	Selects the secondary operating system image for this reboot and sets the configured default operating system image to secondary for future reboots.
<code>serviceos</code>	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the **show images** command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the **primary** or **secondary** optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the **boot set-default** command to change the configured default operating system image.

- If you select **serviceos** as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the **boot system** command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the **boot system** command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

```
show boot-history [all|{vsf member <1-10>}]
```

Description

Shows boot history information. When no parameters are specified, shows the most recent information about the current boot operation, and the three previous boot operations for the switch. When the **all** parameter is specified, the output of this command shows the boot information for the active management module.

.



To view boot-history on a standby, the command must be sent on the conductor console.

Parameter	Description
all	Optional. Shows boot information for the active management module.
vsf member <1-10>	Optional. Display boot history for the specified VSF member

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

The output of this command includes the following information:

Parameter	Description
Index	The position of the boot in the history file. Range: 0 to 3.
Boot ID	A unique ID for the boot . A system-generated 128-bit string.
Current Boot, up for <time>	For the current boot, the show boot-history command shows the number of seconds the module has been running on the current software.
<Timestamp>: boot reason	For previous boot operations, the show boot-history command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values: ▪ <DAEMON-NAME> crash: The daemon identified by <DAEMON-NAME> caused the module to boot. ▪ Kernel crash: The operating system software associated with the module caused the module to boot. ▪ Uncontrolled reboot: The reason for the reboot is not known. ▪ Reboot requested through database: The reboot occurred because of a request made through the CLI or other API.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 2
Boot ID : c34a2c2499004a02bbeeff4992e1fdbd
Current Boot, up for 1 days 13 hrs 13 mins 27 secs

Index : 1
```

```

Boot ID : bfba9bc486304e57904ac717a0ccbdcd
02 Sep 23 02:55:33 : CPU request reset with 0x20201, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:55:33 : Switch boot count is 2

Index : 0
Boot ID : a88a71b7ca9a4574af7e3b811ddfdc7e
02 Sep 23 02:49:26 : Reboot requested by user, Version: FL.10.14.0000-1619-
ga9ec1805bd442~dirty
02 Sep 23 02:50:02 : Switch boot count is 1

Index : 3
Boot ID : f00ba10c8c44457f83fee303d014a89a
25 Aug 23 10:27:42 : Power on reset with 0x1, Version: FL.10.14.0000-1465-
g9df95249d06b0~dirty
25 Aug 23 10:28:18 : Switch boot count is 3
25 Aug 23 10:29:02 : Primary overtemperature fault detected with 0x2 in PSU 1/1

```

Showing the boot history of the active management module and all line modules:

```

switch#
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

```

```

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

```

The following example displays the boot history for the VSF member **2**.

```

switch# show boot-history vsf member 2

Member-2
=====

Index : 0
Boot ID : df99026c194a44f1944a3e7685fb4d90
Current Boot, up for 3 hrs 31 mins 39 secs

Index : 3
Boot ID : 7bf4104903fe4ad1ba4bce40e8099c76
10 Aug 17 10:02:24 : Reboot requested through database
10 Aug 17 10:02:13 : Switch boot count is 2

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: **disabled** (down), **connecting** (establishing connection), **operational** (up), and **unaccessible** (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

External storage commands

address

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}  
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The **no** form of this command deletes an IP address or hostname.

Parameter	Description
<IPV4-ADDR>	Specifies the NAS server IPv4 address, Global.
<IPV6-ADDR>	Specifies the IPv6 address of the NAS server.
<HOSTNAME>	Specifies the hostname of the NAS server. String.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

directory

```
directory <DIRECTORY-NAME>  
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The **no** form of this command clears a directory of an external storage volume.

Parameter	Description
<DIRECTORY-NAME>	Specifies the external storage directory for mapping the volume.

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no directory /home
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage-<VOLUME-NAME>	

disable

disable
no disable

Description

Disables the external storage volume.

The **no** form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage-<VOLUME-NAME>	

enable

enable
no enable

Description

Enables the external storage volume.

The **no** form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage- <i><VOLUME-NAME></i>	

external-storage

```
external-storage <VOLUME-NAME>
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The **no** form of this command deletes an external storage volume.

Examples

Creating the logfiles storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)#
```

Deleting the logfiles storage volume:

```
switch(config)# no external-storage logfiles
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

password (external-storage)

```
password [{plaintext | ciphertext} <PASSWORD>]
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The **no** form of this command clears the password for network attached storage server login.

Parameter	Description
{ciphertext plaintext}	Selects the password format.
<PASSWORD>	Specifies the password. NOTE: When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1
switch(config-external-storage-bak1)# password
Enter the NAS server password: *****
Re-Enter the NAS server password: *****
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

Command History

Release	Modification
10.07 or earlier	--

Command Information


```
switch# show running-config external-storage
```

```
external-storage nfsvol
  address 10.1.1.1
  vrf     nas
  type    nfsv4
  directoty /home
  enable
external-storage scpdev
  address 30.1.1.1
  vrf     nas
  username switchuser
  password ciphertext xxx
  type    scp
  directoty /home
  enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

type

```
type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}
```

Description

Sets the network attached storage access type for reaching the external storage volume. The **no** form of this command deletes an external storage volume.

Parameter	Description
nfsv3	Specifies the NFSv3 network access protocol.
nfsv4	Specifies the NFSv4 network access protocol.
scp	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no type nfsv4
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage- <i><VOLUME-NAME></i>	Administrators or local user group members with execution rights for this command.

username

```
username <USER-NAME>  
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The **no** form of this command clears a username.

Parameter	Description
<i><USER-NAME></i>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no username nasuser
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

vrf

```
vrf <VRF-NAME>
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
<VRF-NAME>	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no vrf nas
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 50, IP-SLA responder 8.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not explicitly configured, the system automatically uses the first DNS server available in its default configuration.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- The configuration of history results is limited to a maximum of 8 IP-SLA sessions. This means that you can enable and store historical performance data, such as response times and availability, for up to 8 individual IP SLA sessions at any given time.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the maximum value when UDP jitter SLA is enabled. Otherwise, 100% packet loss can be seen and UDP jitter SLA probes will result in failure:

```
copp-policy default
  class hypertext priority 6 rate 50000 burst 64
  default-class priority 6 rate 99999 burst 9999
```

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to 'running' state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

https

```
https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

```
no https {get | raw} <URL> [history-interval <HISTORY-INTERVAL>] [cache disable] [name-server {<IPV4-ADDR-DNS-SERVER>|<IPV6-ADDR-DNS-SERVER>}] [probe-interval <PROBE-INTERVAL>] [proxy <PROXY-URL>] [source {<IPV4-ADDR>|<IPV6-ADDR>|IFNAME}] [source-port <SOURCE-PORT-NUM>] [version <VERSION-NUMBER>] [http-raw-request <RAW-PAYLOAD>]
```

Description

Configures HTTPS as the IP-SLA test mechanism. Requires destination URL and type of HTTPS request (get/raw).

The **no** form of this command removes the configuration.



For HTTPS IP-SLA sessions, it is not required to install a certificate on the switch.

Parameter	Description
{get raw}	Selects HTTPS request type as get or raw where the system will generate or provide HTTPS payload.
<URL>	Specifies HTTPS URL address

Parameter	Description
	of syntax. https://<HOST NAME/IP-ADDRESS>:<PORT>/<PATH>.
history-interval <HISTORY-INTERVAL>	Configures the history interval for the IP-SLA. Set the history interval to minimum of two times the probe-interval for the SLA. Range: 60 to 7200.
cache disable	Selects cache option for the HTTPS server. By default the option is enabled.
name-server {<IPV4-ADDR-DNS-SERVER> <IPV6-ADDR-DNS-SERVER>}	Specifies the IPv4 address of DNS server.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 30 to 604800.
proxy <PROXY-URL>	Specifies the probe interval in seconds. Range: 30 to 604800.
source {<IPV4-ADDR> <IPV6-ADDR> IFNAME}	Selects the source, either an IPv4 address, an IPv6 address, or hostname for SLA probes.
source-port <SOURCE-PORT-NUM>	Specifies the value of the source port for the IP-SLA probes.
version <VERSION-NUMBER>	Specifies the source interface to use for sending IP-SLA probes.
https-raw-request <RAW-PAYLOAD>	Specifies the HTTPS raw request. String.

Examples

Configuring HTTPS get with parameters:

```
switch(config-ipsla-1)# https get https://device.arubanetworks.com/root/home.html
```

Configuring HTTPS raw with parameters:

```
switch(config-ipsla-1)# https raw https://device.arubanetworks.com/root/home.html
raw-request "GET /en/US/hmpgs/index.html"
```

Removing the HTTPS raw:

```
switch(config-ipsla-1)# no https raw
https://device.arubanetworks.com/root/home.html raw-request "GET
/en/US/hmpgs/index.html"
```

Command History

Release	Modification
10.16.1000	Added new parameters history-interval . Also, IPv6 addresses can be used.
10.12.1000	Command introduced.

Command Information

Platforms	Command context	Authority
5420	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

ip-sla responder

```
ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip) [<PORT-NUM>]  
[source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf <VRF-NAME>] [ipv6]
```

```
no ip-sla responder <SLA-NAME> (udp-echo | tcp-connect | udp-jitter-voip) [<PORT-NUM>]  
[source {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>|<IFNAME>}] [vrf <VRF-NAME>] [ipv6]
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<SLA-NAME>	Specifies the IP-SLA responder name. Length: 1 to 64 characters.
udp-echo	Enables responder for udp-echo probes.
tcp-connect	Selects TCP connect as the IP-SLA test mechanism.
udp-jitter-voip	Selects VOIP jitter as the IP-SLA test mechanism.
<PORT-NUM>	Specifies the port number to listen for IP-SLA probes. Range: 1 to 65535.
source {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR> <IFNAME>}	Selects the source IPv4 or IPv6 address for SLA probes or the source interface to use for sending IP-SLA probes.
vrf <VRF-NAME>	Specifies the name of the VRF to use.
ipv6	Configures IPv6 responder. This

Parameter	Description
	keyword is required if an IPv6 address is being used by the source interface or VRF. By default, it will be considered as an IPv4 address.

Usage

The IPv6 keyword is required if an IPv6 address is being used by the source interface or VRF. Otherwise, by default, it will be considered as an IPv4 address.

Examples

Configuring IP-SLA responder for udp-echo:

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Configuring IP-SLA responder with IPv6:

```
switch(config)#ip-sla responder SLA1 udp-echo 8000 source 1/1/1 ipv6
```

Configuring IP-SLA responder for udp-jitter-voip:

```
switch(config)#ip-sla responder SLA1 udp-jitter-voip 1025 vrf <VRF>
```

Disabling IP-SLA responder:

```
switch(config)# no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Command History

Release	Modification
10.15	Added ipv6 parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

show ip-sla all

```
show ip-sla all
```

Description

Shows all ip-sla source configuration and status.

Examples

Showing results for ip-sla all:

```
switch# show ip-sla all
SLA Name           : 2 (non-persistent)
Status             : running
SLA Type           : udp-echo
VRF                : default
Source IP          :
Source Interface    :
Domain Name Server  :
Payload Size       : 28
TOS                : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45

IP-SLA session status
IP-SLA Name                : 2 (non-persistent)
IP-SLA Type                : udp-echo
Destination Host Name/IP Address : 10.1.1.2
Destination Port           : 8888
Source IP Address/IFName    :
Source Port                :
Status                     : running

IP-SLA Session Cumulative Counters
Total Probes Transmitted    : 10
Probes Timed-out           : 10
Bind Error                  : 0
Destination Address Unreachable : 0
DNS Resolution Failures     : 0
Reception Error            : 0
Transmission Error         : 0
Operational Status         : down

IP-SLA Latest Probe Results
Last Probe Time            :
Packets Sent               : 1
Packets Received           : 0
Packet Loss in Test        : 100%

Minimum RTT(ms)           :
Maximum RTT(ms)           :
Average RTT(ms)           :
DNS RTT(ms)               :

-----
SLA Name           : echo-udp-sess2 (non-persistent)
Status             : running
SLA Type           : udp-echo
VRF                : default
Source IP          :
Source Interface    :
Domain Name Server  :
Payload Size       : 28
TOS                : 0
Probe Interval(seconds) : 60
History Interval(seconds) : 0
Timeout Interval(seconds) : 45
```

```

IP-SLA session status
IP-SLA Name                : echo-udp-sess2 (non-persistent)
IP-SLA Type                : udp-echo
Destination Host Name/IP Address : 100.1.1.2
Destination Port           : 8888
Source IP Address/IFName    :
Source Port                :
Status                     : running

IP-SLA Session Cumulative Counters
Total Probes Transmitted    : 10
Probes Timed-out           : 0
Bind Error                  : 0
Destination Address Unreachable : 0
DNS Resolution Failures    : 4
Reception Error            : 0
Transmission Error         : 0
Operational Status        : Up

IP-SLA Latest Probe Results
Last Probe Time            :
Packets Sent               : 1
Packets Received           : 1
Packet Loss in Test        : 0.0000%

Minimum RTT(ms)            :
Maximum RTT(ms)            :
Average RTT(ms)            :
DNS RTT(ms)                :

```

Showing results for non-configured ip-sla all:

```

switch# show ip-sla all
IPSLA source is not configured

```

Command History

Release	Modification
10.16.1000	Updated to display history interval .
10.12.1000	Updated to display https as an IP-SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla responder


```
show ip-sla responder <SLA-NAME> [initiator {<SOURCE-IPV4-ADDR>|<SOURCE-IPV6-ADDR>}]
[<SOURCE-PORT-NUM>] [results]
```

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
initiator {<SOURCE-IPV4-ADDR> <SOURCE-IPV6-ADDR>}	Selects the source IPv4 or IPv6 address for SLA probes to use.
<SOURCE-PORT-NUM>	Configures the source port for the IP-SLA test. Range: 1 to 65535.
results	Displays the statistics for a given source IP and port.

Examples

Showing IP-SLA responder configuration:

```
switch# show ip-sla responder SLA3

SLA Name       : SLA3
IP-SLA Type    : Udp-echo
VRF            : Default
Responder Port  : 8000
Responder IP    : 2.2.2.3
Responder Interface : 1/1/1
Responder Status : Running
```

Showing IP-SLA responder with initiator and results parameters:

```
switch# show ip-sla responder SLA1 initiator 2.2.2.1 8000 results

IP-SLA Type      : Udp-echo
VRF Name         : Default
Source IP        : 2.2.2.1
Source Port      : 8000
Responder Port    : 8888
Responder IP      : 2.2.2.3
Responder Interface :
Responder Status  : Running
Packets Received  : 2
Packets Sent      : 2
```

Command History

Release	Modification
10.16.1000	Added new parameters: initiator and results .
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

show ip-sla

```
show ip-sla <SLA-NAME> [{results | history-results}]
```

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
results	Displays the statistics calculated for an SLA type.
history-results	Displays the history statistics calculated for the SLA ID.

Examples

Showing results for ip-sla:

```
switch# show ip-sla xyz results
IP-SLA session status
IP-SLA Name                : xyz
IP-SLA Type                : tcp-connect
Destination Host Name/IP Address: 2.2.2.1
Destination Port           : 8888
Source IP Address/IFName   : 2.2.2.2
Source Port                : 5555
Status                    : running

IP-SLA session cumulative counters
Total Probes Transmitted   : 1
Probes Timed-out          : 0
Bind Error                : 0
Destination Address Unreachable : 0
DNS Resolution Failures   : 0
Reception Error           : 0
Transmission Error        : 0

IP-SLA Latest Probe Results
Last Probe Time           : 2018 Jul 13 02:00:35
Packets Sent              : 1
Packets Received          : 1
Packet Loss in Test       : 0.0000%

Minimum RTT(ms)           : 12
Maximum RTT(ms)           : 12
Average RTT(ms)           : 12
DNS RTT(ms)               : 0
TCP RTT(ms)               : 12
```

Showing history results for ip-sla:

```

switch# sh ip-sla abcd history-results
IP-SLA Name: abcd

Session Details
=====
IP-SLA Type           : tcp-connect      Status                : running

Probe Interval(seconds) : 30              History Interval(seconds) : 600

Source Port           : 3000              Destination Port        : 4000

Source IP Address      : 10.0.0.1
Dest Host Name/IP Address : 10.0.0.2

History Probe Results
=====
Packet Stats
-----
Probes Transmitted      : 4              Packets Sent           : 4
Packets Received        : 4              Loss Percentage         : 0.0000%

Error Stats
-----
Transmission Errors     : 0              Reception Errors       : 0
Bind Errors             : 0              Dest. Unreachable      : 0
Probes Timed-Out        : 0              DNS Resolution Failures : 0

Probe RTT Stats
-----
Min RTT(ms)            : 0              Max RTT(ms)           : 0
Avg RTT(ms)            : 0

DNS RTT Stats
-----
Min RTT(ms)            :                  Max RTT(ms)           :
Avg RTT(ms)            :

```

Command History

Release	Modification
10.16.1000	Added new parameter history-results . Updated to display history interval .
10.12.1000	Updated to display https as an IP-SLA type.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

The speed downshift feature allows the user to link-up at sub-optimal speeds when failing to link-up at the highest advertised speed. There are fixed number of link attempts made to establish link at highest advertised speed and when all of them fail and attempt is made to link-up at a lower possible speed.

This feature requires underlying PHY to have support for the same and hence capability is only added to select set of ports. If a link cannot be established at the highest common denominator within a set number of link attempts, the PHY advertises the next highest speed using auto-negotiation.

Limitations with speed downshift

- Link up may be delayed as certain number of retries are done to establish the link at highest advertise speeds by both link partners before downshifting.
- Link may be established at sub-optimal speed.

L1-100Mbps downshift commands

downshift enable

```
downshift-enable  
no downshift-enable
```

Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

Examples

```
switch(config-if)# interface 1/1/1  
switch(config-if)# downshift-enable
```

Warning: this is a non-standard mode for use only when standards-based auto-negotiation is not able to establish a stable link. Enabling this may cause the port to link at a lower than expected speed and should not be used on ports that are members of a LAG. Support calls may require this feature to be disabled

```
Continue (y/n)?
```

```
switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface  
interface 1/1/1  
    downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1  
switch(config-if)# no downshift-enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

show interface

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical]  
show interface [<IFNNAME>|<IFRANGE>] [extended [non-zero] | [human-readable]]  
show interface [<IFNNAME>] monitor [human-readable]  
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief]  
show interface lag [<LAG-ID>] [extended [non-zero] | [human-readable]]  
show interface lag [<LAG-ID>] monitor [human-readable]
```

Description

Shows active configurations and operational status information for interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
brief	Shows brief info in tabular format.
physical	Shows the physical connection info in tabular format.
extended	Shows additional statistics, including the tx filtered and rx filtered counters. ▪ Rx filter packets are protocol packets received when the protocol is disabled on the switch and there is only one port in the VLAN. Protocols include OSPF, PIM, RIP, LACP, and LLDP. ▪ An example of a Tx filtered packet would be a multicast packet being filtered from going out of the ingress port.

Parameter	Description
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G. This is available only in the CLI interface output.
non-zero	Shows only non zero statistics.
LAG	Shows LAG interface information.
monitor	Continuously monitor interface statistics.
LOOPBACK	Shows loopback interface information.
TUNNEL	Shows tunnel interface information.
VLAN	Shows VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1

Examples

Showing information when interface 1/1/1 is configured:

```
MDI mode: MDIX
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: all
Rate collection interval: 300 seconds
```

Rates	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00

Statistics	RX	TX	Total
Packets	0	0	0
Unicast	0	0	0
Multicast	0	0	0
Broadcast	0	0	0
Bytes	0	0	0
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0

Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information when the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

Showing information when the interface is currently linked with energy-efficient-ethernet negotiated:

```
switch(config-if)# show interface 1/1/1
Interface 1/1/1 is up
...
Energy-Efficient Ethernet is enabled and active
```

Showing information when the interface is configured with EEE and the EEE has auto-negotiated:

```
switch(config-if)# show interface 1/1/1 physical
```

EEE	PoE Power	Link	Admin	Speed	Flow-Control
Port	Type	Status	Config	Port	Status Config
Status Config	(Watts)	State	Information	Description	
1/1/1	1GbT	up	up	1G	off off
on on	--	10M/100M/1G	--	auto	

Showing the monitor information:



In monitor mode, the CLI refreshes data automatically until it is exited by entering **q**. Pressing **?** opens the help menu to display which options are available in this context.

```
Interface 1/1/1 is up
```

Rate	RX	TX	Total (RX+TX)
-----	-----	-----	-----
MBits / sec	30196.43	30196.43	60392.85
MPkts / sec	58977.39	58977.40	117954.79
Unicast	0.00	0.00	0.00
Multicast	58977.39	58977.40	117954.79
Broadcast	0.00	0.00	0.00
Utilization %	75.49	75.49	150.98
Statistic	RX	TX	Total (RX+TX)
-----	-----	-----	-----
Packets	4756527649	4756527865	9513055514
Unicast	0	0	0

```

Multicast          4756527649          4756527865          9513055514
Broadcast          2                    0                    2
Bytes              304417778668         304417795428         608835574096
Jumbos             0                    0                    0
Dropped            0                    19028847730          19028847730
Pause Frames       0                    0                    0
Errors             0                    0                    0
CRC/FCS            0                    n/a                  0
help: ?, quit: q

```

```

Help for Interface Monitor
h  Toggle human-readable mode
c  Clear interface statistics
Does not apply to rates
Arrows, PgUp, PgDn, Home, End
Navigate interface statistics
Delay: 2
help: ?, quit: q

```

Showing the output for interface 1/1/1 in human-readable format:



In human-readable format, the **< 1** symbol for **Utilization** indicates that the amount of packets is between zero and one. This is true in cases where the number of bytes increases but the number of packets and the **Utilization** value is not displayed even in the normal output, where the human-readable parameter is not included in the command.

```

switch(config-if)# show interface 1/1/1 human-readable
Interface 1/1/1 is up

```

Rate	RX	TX	Total (RX+TX)
Bits / sec	3M	3M	6M
Pkts / sec	316	316	633
Unicast	319	319	638
Multicast	0	0	0
Broadcast	0	0	0
Utilization %	< 1	< 1	< 1
Statistic	RX	TX	Total
Packets	577K	577K	1M
Unicast	577K	577K	1M
Multicast	0	51	51
Broadcast	0	15	15
Bytes	744M	745M	1G
Jumbos	0	0	0
Dropped	0	0	0
Filtered	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Showing information about extended counters:



The output of the `show interface extended` command varies depending on the switch model and configuration.

```
switch(config-if)# show interface 1/1/17 extended
-----
Interface 1/1/17
-----
Statistics                                         Value
-----
Dot1d Tp Port In Frames                          547
Dot1d Tp Port Out Frames                         608
Dot3 In Pause Frames                             0
Dot3 Out Pause Frames                           0
Ethernet Stats Broadcast Packets                 19
Ethernet Stats Bytes                             40162
Ethernet Stats Packets                           342
...
-----
Error-Statistics                                 Value
-----
Dot1d Base Port MTU Exceeded Discards            0
Dot3 Control In Unknown Opcodes                 0
Dot3 Stats Alignment Errors                     0
Dot3 Stats FCS Errors                           0
Dot3 Stats Frame Too Longs                      0
Dot3 Stats Internal Mac Transmit Errors          0
Ethernet RX Oversize Packets                     0
...
```

Showing interface link-status:

```
switch# show interface link-status
-----

Port          Type          Physical Link      Last
Link State   Transitions Change
-----
1/1/1         1G-BT         down      0          --
1/1/2         1G-BT         up        1          1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/3         1G-BT         up        1          1 minute ago (Fri Mar 09
12:36:56 UTC 2018)
1/1/4         --            down      0          --
1/1/5         --            down      0          --
```

Showing interface loopback 1 link-status:

```
-----

Port          Type          Physical Link      Last
Link State   Transitions Change
-----
loopback1     --            up        --         --
```

Showing interface 1/1/2-1/1/3 link-status:

Port	Type	Physical Link State	Link Transitions	Last Change
1/1/2	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3	1G-BT	up	1	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)

Showing interface link-status:

```
switch# show interface link-status
```

Port	Type	Physical Link State	Link Transitions	Link Flaps Ignored	Last Change
1/1/1	1G-BT	down	0	0	--
1/1/2	1G-BT	up	1	0	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/3	1G-BT	up	1	0	1 minute ago (Fri Mar 09 12:36:56 UTC 2018)
1/1/4	--	down	0	0	--
1/1/5	--	down	0	0	--

Showing state information when interface is blocked:

```
8360(config-if)# show interface 1/1/1
```

Interface 1/1/1 is up (Blocked)
Admin state is up
State information: Blocked by UDLD
Link state: up for 1 minute (since Mon Jun 10 09:25:27 UTC 2024)
Link transitions: 1
Description:
Persona:
Hardware: Ethernet, MAC Address: 00:fd:45:67:85:91
MTU 1500
Type 10G-LR / 10G SFP+ LR
Full-duplex
qos trust none
Speed 10000 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: access
Access VLAN: 1
Rate collection interval: 300 seconds

Rate	RX	TX	Total (RX+TX)
Mbits / sec	0.00	0.00	0.00
KPkts / sec	0.00	0.00	0.00
Unicast	0.00	0.00	0.00
Multicast	0.00	0.00	0.00
Broadcast	0.00	0.00	0.00
Utilization %	0.00	0.00	0.00
Statistic	RX	TX	Total
Packets	15	15	30

Unicast	12	12	24
Multicast	3	3	6
Broadcast	0	0	0
Bytes	1350	1350	2700
Jumbos	0	0	0
Dropped	0	0	0
Pause Frames	0	0	0
Errors	0	0	0
CRC/FCS	0	n/a	0
Collision	n/a	0	0
Runts	0	n/a	0
Giants	0	n/a	0

Command History

Release	Modification
10.15	Added state information when port goes into down state.
10.11	Added <code>monitor</code> parameter.
10.10	Added <code>human-readable</code> parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show interface statistics

```

show interface [<IFNAME>|<IFRANGE>] statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] statistics monitor [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics [non-zero] [human-readable]
show interface [<IFNAME>|<IFRANGE>] error-statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] statistics monitor [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics [non-zero] [human-readable]
show interface lag [<LAG-ID>] error-statistics monitor [non-zero] [human-readable]
show interface vxlan <VXLAN-ID> statistics [non-zero] [human-readable]

```

Description

Shows statistics for switch interfaces such as packets transmitted and received, bytes transmitted and received, broadcast and multicast packets.

Parameter	Description
<code><IFNAME></code>	Specifies a interface name.
<code><IFRANGE></code>	Specifies the port identifier range.

Parameter	Description
LAG	Shows LAG interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256
VXLAN	Shows the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1
monitor	Continuously monitor interface statistics.
human-readable	Shows statistics rounded to the nearest power of 1000, for example, 1K, 345M, 2G.
non-zero	Shows only non zero statistics.

Examples

Showing statistics of all interfaces:

Showing statistics of all interfaces with only non-zero statistics:

Showing statistics of all interfaces in the human-readable format:

Showing statistics of a single interfaces:

Showing statistics of all members of a LAG interface:

Showing error statistics of all interfaces:

Showing monitor statistics:



The rows and columns of show interface monitor statistics depends on the length of width of the client terminal. The CLI can be navigated using the arrow keys as well as the PageUp, PageDown, Home, and End keys.

```
show interface statistics monitor
```

```
-----
Interface                RX Bytes          RX Packets >>
-----
1/1/1                    3440525421984      53758209526
1/1/2                    3440526607008      53758228042
1/1/3                    3440527785312      53758246453
1/1/30                   3440559671264      53758744653
1/1/31                   3440560851680      53758763098
1/1/32                   3440562028704      53758781489
```

```
help: ?, quit: q
```

Help for Interface Monitor

f Toggle full statistics
h Toggle human-readable mode
n Toggle non-zero mode
r Toggle rate display

c Clear interface statistics
Does not apply to rates

Arrows, PgUp, PgDn, Home, End
Navigate interface statistics

Delay:2

help: ?, qui

Showing monitor error statistics in human-readable format:

```
show interface 1/1/1-1/1/3,1/1/30-1/1/32 error-statistics monitor human-readable
```

Interface	RX Errors	TX Errors	RX Giants	RX Runts	CRC/FCS	Collisions
1/1/1	0	0	0	0	0	0
1/1/2	0	0	0	0	0	0
1/1/3	0	0	0	0	0	0
1/1/30	0	0	0	0	0	0
1/1/31	0	0	0	0	0	0
1/1/32	0	0	0	0	0	0

Human-readable

help: ?, quit: q

Help for Interface Monitor

h Toggle human-readable mode
n Toggle non-zero mode

c Clear interface statistics
Does not apply to rates

Arrows, PgUp, PgDn, Home, End
Navigate interface statistics

Delay:2

help: ?, quit: q

Command History

Release	Modification
10.11	Added moitor parameter.
10.10	Added human-readable parameter.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show interface downshift-enable

show interface [<IFNAME>|<IFRANGE>] downshift-enable

Description

Displays speed downshift information, including the interface speed status and configuration.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.

Examples

Showing automatic downshift information:

```
switch(config-if)# show interface downshift-enable
-----
Port          Downshift          Speed
             Enabled | Active   Status   | Config
-----
1/1/1         yes       yes     100M-FDx  auto
1/1/2         yes       no      1G        auto
1/1/3         yes       no     100M-FDx 100M-FDx
1/1/4         no        no      --        auto
```

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable
-----
Port          Downshift          Speed
             Enabled | Active   Status   | Config
-----
1/1/2         yes       no      1G        auto
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	

show running-config interface

```
show running-config interface [<IFNAME>|<IFRANGE>]  
show running-config interface [lag | loopback | tunnel | vlan ] [<ID>]
```

Description

Displays active configurations of various switch interfaces.

Parameter	Description
<IFNAME>	Specifies a interface name.
<IFRANGE>	Specifies the port identifier range.
LAG	Specifies LAG interface information
LOOPBACK	Specifies loopback interface information.
TUNNEL	Specifies tunnel interface information.
VLAN	Specifies VLAN interface information.
<LAG-ID>	Specifies the LAG number. Range: 1-256.
<LOOPBACK-ID>	Specifies the LOOPBACK number. Range: 0-255.
<TUNNEL-ID>	Specifies the tunnel ID. Range: 1-255.
<VLAN-ID>	Specifies the VLAN ID. Range: 1-4094.
VXLAN	Specifies the VXLAN interface information.
<VXLAN-ID>	Specifies the VXLAN interface identifier. Default: 1.

Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2  
  
interface 1/1/2  
  no shutdown  
  description DC-23  
  exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback  
  
interface loopback 1  
  description lb interface 1  
  exit  
interface loopback 2  
  description lb interface 2  
  exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback  
  
No loopback interfaces configured.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

Mirror statistics

Mirror statistics are reset for a Mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the Mirror session and during a failover.

Classifier policies and mirroring sessions

Network traffic can be mirrored to a destination interface in two ways:

- Using a mirroring session alone.
- Using Classifier Policies with mirror actions in conjunction with a mirroring session.

Basic mirroring sessions provide coarse control over the type of traffic mirrored from a source: all received, all transmitted, or both. However, a traffic class within a Classifier Policy applied to a source can provide much finer grained control of mirrored traffic. For example, a policy can match on many different aspects of the Ethernet or IPv4 or IPv6 header information in each frame or packet received or transmitted on an interface.

The steps to configure a policy and class with a mirror action are the following:

1. Configuring a mirroring session with a destination interface.
2. Enabling the mirroring session.
3. Configuring the Classifier Policy, specifying the mirroring session ID in the mirror action.

If the packets being mirrored are received from a VLAN that is not allowed on the mirror destination, the mirrored packets would be dropped at the mirror destination interface. When the mirrored packets are dropped at the destination, the mirror output packet and byte count will increment, however the packets will not be received at the mirror destination.

The mirror destination port among the active mirror sessions must be unique. That is, if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.

VLAN as a source

AOS-CX allows configuration of VLAN as a mirroring source. When a VLAN source is configured in the 'rx' direction, all packets are mirrored as they are received in the switch. When a VLAN source is configured in 'tx' direction, all packets are mirrored as they are transmitted out of the switch.

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs in each direction of a given mirror session.

Same VLANs can be configured as a mirror source for multiple sessions.

Note: When changing a source VLAN in an enabled mirror session (that is, adding, changing direction, or removing), mirrored packets being transmitted out the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways:

1. Reenter the `source vlan` command with the new preferred direction.
2. Use the `no` form of the command with a direction (rx or tx) to selectively remove the specified direction. Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

For packets bridged through the switch:

If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in the 'rx' direction, packets are mirrored in the pre-routed form with the destination MAC address as the switch address.
- If the mirror is configured in the 'tx' direction, packets are mirrored in the post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in the 'both' direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behaviors for mirroring with VLAN as source:

- If the mirror is configured in the 'rx' or 'tx' direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the 'both' direction, two copies of the packets are mirrored to the mirror destination.

Mirroring commands

clear mirror

```
clear mirror [all | <SESSION-ID>]
```

Description

Clears the mirror statistics for all configured mirror sessions or a specified session

Parameter	Description
all	Specifies all configured sessions.
<SESSION-ID>	Specifies a numeric identifier for the session. Range: 1 to 4

Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

clear mirror endpoint

```
clear mirror endpoint [<NAME>]
```

Description

Clears mirror endpoint statistics for all configured mirror endpoints. The optional parameter can be added to clear a specific mirror endpoint.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be cleared.

Examples

Clearing statistics for all configured mirror endpoints:

```
switch# clear mirror endpoint
```

Clearing mirror statistics for mirror endpoint test:

```
switch# clear mirror endpoint test
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

comment

```
comment <COMMENT>  
no comment
```

Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The **no** form of this command removes the comment.

Parameter	Description
<COMMENT>	A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods.

Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3)# comment This Mirror will be removed during next  
maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3)# no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test)# comment Monitor statistics on each endpoint interfaces
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID> config-mirror-endpoint	Administrators or local user group members with execution rights for this command.

copy tcpdump-pcap

```
copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>
```

Description

Saves packet capture files to external storage.

Parameter	Description
<FILE-NAME>	Specifies the packet capture file to save.
<REMOTE-URL>	Specifies the external storage to which the packet capture file will be saved.

Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using **diag utilities list-files**.

Examples

Saving my_capture_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's passwd:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap                               100%   156   219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

copy tshark-pcap

`copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]`

Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

Parameter	Description
<code><REMOTE-URL></code>	Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is: {tftp:// sftp://<USER>@} {<IP> <HOST>} [:<PORT>] [;blocksize=<SIZE>]/<FILE>
<code>vrf <VRF-NAME></code>	Specifies the name of a VRF. Default: default.

Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap

root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap                               100% 156   219.8KB/s   00:00
Copied successfully.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

destination cpu

destination cpu
no destination cpu

Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the **Supportability Guide**.

The **no** form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

destination interface

destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}

Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

Parameter	Description
<INTERFACE-ID>	Specifies a interface. Format: member/slot/port.
<LAG-NAME>	Specifies a LAG (link aggregation group) identifier.

Usage

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

Switch	Destination interface limit per mirror session (4 possible sessions)
5420	64

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

destination tunnel

```
destination tunnel <TUNNEL-IPV4> source <SOURCE-IPv4-ADDR>
dscp <DSCP-VALUE> vrf <VRF-NAME>
no destination tunnel
```

Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror_err_tunnel_oob_port_not_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel_route_resolution_not_populated" operation status.



The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

The **no** form of this command will cease the use of the tunnel and disable the session.

Parameter	Description
<TUNNEL-IPv4-ADDR>	Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<SOURCE-IPv4-ADDR>	Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.
<DSCP-VALUE>	Specifies the DSCP value to be carried within the DS field of ERSPAN packet header. Range: 0 to 63. Default: 0.
<VRF-NAME>	Specifies a VRF name. Default: default.

Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf
default
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

diagnostic

diagnostic

```
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the **delete-file** form of this command to delete the most recent capture file.

Since **file** and **delete-file** are optional, the behavior of the base command **diag utilities tshark** does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

Parameter	Description
file	Saves captured packets to a temporary file.
delete-file	Deletes the most recent captured file.

Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

diag utilities tcpdump

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-files |  
vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-ADDR> | source-ip  
<IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port <PORT> |  
destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-type <ETH-NUM>]
```

Description

Captures traffic received or transmitted over a network.

Parameter	Description
command <TEXT>	Captures packets based on a specified tcpdump command string.
delete file <FILE-NAME>	Deletes specified tcpdump list files.
list-files	Lists all the tcpdump capture files saved on the device.
vrf <VRF-NAME>	Captures packets on the specified VRF. If no VRF is named, the default is used.
count <COUNT-NUM>	Runs the tcpdump command until the specified number of packets are captured. Range: 1-2147483647.
proto <PROTO-NUM>	Captures packets of a particular type based on IP protocol number. Range: 0-255.
host-ip <IP-ADDR>	Captures packets matching with the source or destination IP address.
source-ip <IP-ADDR>	Captures packets from the specified IP address.
destination-ip <IP-ADDR>	Captures packets sent to the specified IP address.
host-port <PORT>	Captures packets matching with the source or destination port.
source-port <PORT>	Captures packets from the specified IP port.
destination-port <PORT>	Captures packets sent to the specified IP port.
verbosity <LEVEL>	Captures packets of the specified verbosity. Range: level1-level4. If no verbosity is specified, the default is level1.
print <DATA>	Captures the data of each packet. The maximum is 262144 bytes
ethernet-type <ETH-NUM>	Captures packets based on the particular ethernet type. Range: 0-65535.

Usage

- When using the **command** option, the only traffic captured will be packets that have been mirrored to the CPU.
- When using the **command** option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
 - -i/--interface
 - -Z
 - -B/--buffer-size
 - -C
 - -W
 - -Z/--relinquish privileges
- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using **command -r** to read a file, do not provide any directory path characters. Use list-files command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the **copy tcpdump-pcap** command.

Examples

Inspecting traffic mirrored to the CPU via tcpdump and saving the output to my_capture_file.pcap:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
my_capture_file.pcap
```

Reading my_capture_file.pcap:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB (Ethernet)
 1 11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved, length
12
    0x0000: 0000 0304 0006 0000 0000 0000 0000 0000 86dd .....
    0x0010: 600a 7e47 0014 1140 0000 0000 0000 0000 `~G...@.....
    0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 .....
    0x0030: 0000 0000 0000 0001 9d7e 007b 0014 0027 .....~.{...
    0x0040: 1601 0001 0000 0000 0000 0000 .....
 2 11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved, length
12
    0x0000: 0000 0304 0006 0000 0000 0000 0000 0000 86dd .....
    0x0010: 6b8d 23c5 0014 1140 0000 0000 0000 0000 k.#....@.....
    0x0020: 0000 0000 0000 0001 0000 0000 0000 0000 .....
    0x0030: 0000 0000 0000 0001 007b 9d7e 0014 0027 .....{.~....'
```

```
0x0040:  d681 0001 c016 0000 0000 0000
```

Removing my_capture_file.pcap:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
Successfully removed file
```

Command History

Release	Modification
10.08	Command introduced

Command Information

Platforms	Command context	Authority
5420	Manager (#)	Administrators or local user group members with execution rights for this command.

disable

disable

Description

Disables the mirroring session specified by the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the **show mirror** command for that session ID shows an **Admin Status** of **disable** and an **Operation Status** of **disabled**.

Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

enable

enable

Description

Enables the mirroring session for the current command context.

Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the **show mirror** command for that session ID shows an **Admin Status** of **enable** and an **Operation Status** of **enabled**.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of **rx** or **both**):

- The attempt to enable the mirroring session fails and an error is returned.



When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config-mirror-<SESSION-ID>	Administrators or local user group members with execution rights for this command.

mirror session

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The **no** form of this command removes an existing mirroring session from the configuration.

Parameter	Description
<code><SESSION-ID></code>	Specifies the session identifier. Range: 1 to 4

Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

mirror endpoint

```
mirror endpoint <NAME>
no mirror endpoint <NAME>
```

Description

Creates the specified mirror endpoint or enters its context if it already exists. The specifics of a mirror endpoint are created or altered while in the mirror endpoint context and the mirror endpoint is enabled or disabled from this context. It may be possible to support different encapsulations by different ASICs. For example, UDP for PVOS compatibility. Termination of GRE encapsulation is also supported.

The **no** form of this command removes an existing mirror endpoint. An enabled mirror endpoint is automatically disabled first before removal.

Parameter	Description
<code><NAME></code>	Specifies mirror endpoint name.

Examples

Creating a mirror endpoint named test :

```
switch(config)# mirror endpoint test
```

Deleting mirror endpoint named test:

```
switch(config)# no mirror endpoint test
```

Configuring a mirror endpoint named test :

```
6100(config)# mirror endpoint test
6100(config-mirror-endpoint-test)#
6100(config-mirror-endpoint-test)# destination
    interface Specify interfaces to send traffic
6100(config-mirror-endpoint-test)# destination interface
    IFNAMELIST An interface, a range or a comma seperated list of interfaces
6100(config-mirror-endpoint-test)# destination interface 1/1/3
    <cr>
6100(config-mirror-endpoint-test)# destination interface 1/1/3
6100(config-mirror-endpoint-test)#
6100(config-mirror-endpoint-test)# source 1.1.1.1 destination 1.1.1.2 id 1 vrf
default
6100(config-mirror-endpoint-test)#
```



Only physical ports can be configured as interface for mirror-endpoint destination. LAG port is not supported as interface for mirror-endpoint destination.



The maximum allowed number of destination interfaces for both mirror-session and mirror-endpoint is 1.

Command History

Release	Modification
10.13.1000	Added support for 4100i, 6000, and 6100 switches.
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

show mirror

```
show mirror [<SESSION-ID>]
```

Description

Shows information about mirroring sessions. If `<SESSION-ID>` is not specified, then the command shows a summary of all configured mirroring sessions. If `<SESSION-ID>` is specified, then the command shows detailed information about the specified mirroring session.

Parameter	Description
<code><SESSION-ID></code>	Specifies the session identifier. Range: 1 to 4

Usage

Information in the **Admin Status** column of the command output indicates the configured status. The admin status is one of the following values:

- **enable**: The mirroring session is enabled.
- **disable**: The mirroring session has been configured but not yet enabled, or has been disabled.

Information in the **Operation Status** column indicates the status of the mirroring session. Operation status is one of the following values:

- **dest_doesnt_exist**: The configured destination interface is not found in the system. The mirroring session cannot be enabled.
- **destination_shutdown**: **The mirroring session is enabled, but the destination interface is shut down. No traffic** can be monitored.
- **disabled**: The mirroring session is disabled and is not in an error condition.
- **enabled**: The mirroring session is enabled.
- **external/driver_error**: An internal ASIC hardware error occurred.
- **hit_active_sessions_capacity**: The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.
- **internal_error**: An invalid parameter was passed to the ASIC software layer.
- **no_dest_configured**: The mirroring session does not have a destination interface configured.
- **no_name_configured**: A software error occurred. The mirroring session does not have a session ID in its configuration.
- **null_mirror**: A software error occurred. The session object reference is invalid.
- **out_of_memory**: The system is out of memory, reboot recommended.
- **tunnel_route_resolution_not_populated**: If the destination tunnel IP address is not reachable.
- **unknown_error**: An unexpected error occurred.

Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID  Admin Status  Operation Status
---  -
1   enable       enabled
2   disable      disabled
3   disable      disabled
4   enable       internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	

show mirror endpoint

show mirror endpoint [<NAME>]

Description

Shows a list of all configured mirror endpoints, their Admin Status and their Operation Status. The optional parameter will display the details of the specified mirror endpoint if it exists.

Parameter	Description
<NAME>	Specifies name of the mirror endpoint instance to be displayed.

Examples

Showing a summary of all configured mirror endpoints on the switch:

```
switch# show mirror endpoint
Name      Admin Status  Operation Status
-----
test      enable        enabled
monitor   disable       disabled
```

Showing the details of enabled mirror endpoint test:

```
switch# show mirror endpoint test
Mirror Endpoint: audit
Admin Status: enable
Operation Status: enabled
Comment: Mirror Endpoint Audit
Type: gre
```

```
Tunnel: source 1.1.1.1 destination 1.1.1.2 id 1 vrf default
Interface: 1/1/3
Output Packets: 123456789
Output Bytes: 0
```



"Output Packets" in "show mirror endpoint [name]" is only supported for statistics.

"Output Bytes" in "show mirror endpoint [name]" is not supported due to ASIC limitation.

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

shutdown

shutdown
no shutdown

Description

Enables mirror endpoint from its default disabled state. To verify the mirror endpoint was successfully activated, run the **show mirror endpoint NAME** command and verify that the **Admin Status** and **Operational Status** has changed from disabled to enabled. If the status value remains disabled, consult the system logs to determine the reason for activation failure. To disable the mirror endpoint, first disable the remote mirror session on the switch that's originating the data. Next, use the `shutdown` command to disable the mirror endpoint.

Examples

Enabling a mirror endpoint:

```
switch(config)# mirror endpoint test
switch(config-mirror-endpoint-test)# no shutdown
```

Disabling a mirror endpoint:

```
switch(config)# mirror endpoint test
```

```
switch(config-mirror-endpoint-test) # shutdown
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

source

```
source <SOURCE-IP> destination <DESTINATION-IP> id <1-4294967295> [vrf <VRF_NAME>] [type {gre}]  
no source
```

Description

Configures tunnel parameters of the mirror endpoint. Configuring a tunnel parameter to a mirror endpoint will replace the existing configuration. By default the VRF is **default**, users can also explicitly provide a custom VRF. The default tunnel type is considered to be GRE and users also have the option to explicitly give type as GRE.

The **no** form removes the tunnel parameters of the mirror endpoint.

Parameter	Description
<SOURCE-IP>	Specifies L3 encapsulated IPv4 source in the form A.B.C.D.
<DESTINATION-IP>	Specifies L3 encapsulated IPv4 destination in the form A.B.C.D.
id	Specifies tunnel identifier from the encapsulated packet.
<VRF_NAME>	Specifies the name of VRF for which the tunnel belongs to.

Examples

Configuring a tunnel parameter to a mirror endpoint:

```
switch(config-mirror-endpoint-test) # source 1.1.1.1 destination 7.7.7.7 id 1 vrf  
default type gre
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

source interface

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]  
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored. The **no** form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

Parameter	Description
<PORT-NUM>	Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).
<LAG-NAME>	Specifies the identifier for the LAG (link aggregation group).
<DIRECTION>	Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following:
both	Mirror both transmitted and received packets.
rx	Mirror only received packets.
tx	Mirror only transmitted packets.

Usage

There is a limit of source interfaces in each direction of a given mirror session:

Switch	Source interface limit per mirror session (4 possible sessions)
5420	64

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination.



When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1) # source interface  
LAG-NAME      Enter a LAG name. For example, lag10  
PORT-NUM      Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config) # mirror session 1  
switch(config-mirror-1) # source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config) # mirror session 2  
switch(config-mirror-2) # source interface 1/1/3 tx  
switch(config-mirror-2) # source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2) # no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3) # source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1) # source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4) # source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4) # no source interface lag1 rx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config-mirror-<SESSION-ID></code>	Administrators or local user group members with execution rights for this command.

source vlan

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> {rx | tx | both}
```

Description

Mirroring with VLAN as a source is supported in the following traffic directions:

- **both** - traffic received and transmitted
- **rx** - only received traffic
- **tx** - only transmitted traffic

More than one source VLAN can be configured in a mirror session. Each such VLAN may specify its own direction.

There is a limit of 1024 source VLANs for a given mirror session. There is also a limit of 4096 source VLANs across all mirror sessions.



When changing a source VLAN in an enabled mirror session (i.e. adding, changing direction, or removing) mirrored packets being transmitted out of the mirror destination port from other mirror sources may be briefly interrupted during the reconfiguration.

Direction of an existing source VLAN can be updated in one of two ways.

- Reenter the **source vlan <VLAN-NUM> <direction>** command with the new preferred direction.
- Use the **no source vlan <VLAN-NUM> <direction>** form of the command with a direction (**rx** or **tx**) to selectively remove the specified direction.

Specifying the last remaining direction for that VLAN will remove the VLAN from the configuration entirely.

Mirroring allows configuration of VLAN as a source. When VLAN source is configured in the **rx** direction, all packets are mirrored as they are received in the switch. When VLAN source is configured in **tx** direction, all packets are mirrored as they are transmitted out of the switch.

For packets bridged through the switch:

- If the mirror is configured in 'both' direction, two copies of packets are mirrored, otherwise one copy of the packet will be mirrored.

For routed packets:

- If the mirror is configured in **rx** direction, packets are mirrored in the pre-routed form with the Destination MAC address as the switch address.
- If the mirror is configured in **tx** direction, packets are mirrored in post-routed form with the source MAC as the switch address. Destination MAC is the nexthop gateway or station.
- If the mirror is configured in **both** direction, one copy of the packet will be mirrored.

Control plane packets generated by the switch's CPU are processed both in the ingress and the egress packet processing pipeline. The following are the behavior for mirroring with VLAN as source:

- If the mirror is configured in the **rx** or **tx** direction, the packets are mirrored to the mirror destination.
- If the mirror is configured in the **both** direction, two copies of the packets are mirrored to the mirror destination.

The **no** form command will cease mirroring traffic from the specified source VLAN and remove the source from the mirror configuration.

Parameter	Description
VLAN-NUM	Selects the VLAN number.
direction	Specifies the direction of mirroring. tx (transmit), rx (receive), or both .

Examples

Creating a mirror session and adding a VLAN as a source of traffic in both directions on that port:

```
switch# configure terminal
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Creating a mirror session and adding two VLANs as sources of traffic:
directions:

```
switch# configure terminal
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Configuring the source in session 2 to receive by specifying the source interface configuration:

```
switch(config-mirror-2)# source vlan 10 rx
```

Removing the first source interface in session 2 entirely, and removing the transmit direction from the other so that mirroring only occurs in the receive direction:

```
switch(config-mirror-2)# source vlan 10 rx
switch(config-mirror-2)# source vlan 20 tx
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	<code>config</code>	Administrators or local user group members with execution rights for this command.

Chapter 10

Monitoring a device using SNMP

Configuring SNMP: Refer to the *SNMP/MIB Guide* for information on how to add SNMP so a device can be monitored from a network management system (NMS).

Configuring an SNMP trap receiver: Refer to the *SNMP/MIB Guide* and specific information about the `show snmp trap` command to enable SNMP traps.

Chapter 11

Power-over-Ethernet

- The Power-over-Ethernet (PoE) subsystem manages power supplied to devices using standard Ethernet data cables. A Power Sourcing Equipment (PSE) supplies DC power as well as Ethernet connectivity to a Powered Device (PD) using a standard Ethernet cable. The maximum current depends on the PD Requested Class.
- A PoE subsystem contains two parts : a PSE and PD. A Power Sourcing Equipment (PSE) is a device that provides power through a standard Ethernet cable. A PoE capable switch functions as PSE. All Aruba PoE switches are considered as PSEs. A PD is a device powered by a PSE. Examples of PD are VoIP phones, Wireless APs, and IP cameras.
- When a PD or any network cable is connected to a PSE port, the PSE applies a detection voltage and measures the resistance value of the PD. If resistance is within IEEE 802.3 standard values (23 - 26k ohm), the connected device is treated as PD and classification begins. For legacy devices to be detected, you must enable prestandard detection on the switch.
- PDs are divided into different types and classes based on PD power requirements. The power supplied by the PSE is higher than the power PD draws to accommodate for the line losses that can result with the use of the standard maximum length cable(100m).
 - Type 1: PSE can supply maximum of 15.4W, and PD can draw a maximum of 13W.
 - Type 2: PSE can supply maximum of 30W, and PD can draw a maximum of 25.5W.
 - Type 3: PSE can supply maximum of 60W, and PD can draw a maximum of 51W.
 - Type 4: PSE can supply maximum of 90W, and PD can draw a maximum of 71W.
- Classes of PD:
 - Class 0: Type1 PD, it can draw a maximum of 13W.
 - Class 1: Type1 PD, it can draw a maximum of 3.84W.
 - Class 2: Type1 PD, it can draw a maximum of 6.49W.
 - Class 3: Type1 PD, it can draw a maximum of 13W.
 - Class 4: Type2 PD, it can draw a maximum of 25.5W.
 - Class 5: Type3 PD, it can draw a maximum of 40W.
 - Class 6: Type3 PD, it can draw a maximum of 51W.
 - Class 7: Type4 PD, it can draw a maximum of 62W.
 - Class 8: Type4 PD, it can draw a maximum of 71.3W.
- IEEE 802.3bt introduced 4-Pair PoE as a means of supplying higher power to PDs that need more than the current 25.5W supplied by IEEE 802.3at. To increase the available power without damaging the Ethernet cable, the standard introduced the ability to use all four pairs within the Ethernet cable instead of the two pairs used by previous standards (802.3at, 802.3af).
- Supported protocols:
 - Compatibility with IEEE 802.3af, 802.3at, 802.3bt and prestandard.
 - Long first class event supported on Type 3-4 PSE.
 - Support for Single Signature (SS) Type 0-6 and Dual Signature (DS) Type 0-4 PDs.
 - Multi-Event classification permits mutual ID of SS Class 0-6 and DS Class 0-4.

- Support LLDP Data Link Layer (DLL) Type 1-2 extension 12-octet TLV and Type 3-4 extension 29-octet TLV.
- Default PSE assigned class delivers the maximum PSE capable power at initial power up based on PD requested class.
- Always-on PoE is a feature that provides the ability for a switch to continue to provide power across user initiated reboots through software. Always-on PoE is enabled by default and no additional configuration is needed.



PDs only remain powered, no data transfer or PoE power negotiation can occur until the switch has completely booted up and in normal operation. PD faults occurring prior to full switch boot up will result in PoE power removal and restart the detection process only after switch returns to normal operation.

PoE commands

All PoE configuration commands except **quick-poe**, **threshold configuration** and **always-on poe configuration** are entered at the **config-if** context. The PoE threshold command is used at the system level whereas the **always-on poe** and **power-over-ethernet quick-poe** commands are set at the slot level. These commands can only be configured in the global configuration context.

lldp dot3 poe

```
lldp dot3 poe
no lldp dot3 poe
```

Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The **no** form of this command disables 802.3 TLV list in LLDP.

Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

lldp med poe

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The **no** form of this command disables MED TLV list in LLDP.

Parameter	Description
[priority-override]	System defined name of the interface.

Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet

```
power-over-ethernet
no power-over-ethernet
```

Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module. The **no** form of this command disables per-interface power distribution.

Examples

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1  
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1  
switch(config)# no power-over-ethernet  
Interface PoE cannot be disabled when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet allocate-by

```
power-over-ethernet allocate-by {usage | class}  
no power-over-ethernet allocate-by {usage | class}
```

Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The power allocation method can be changed on an interface through port-access (User roles or RADIUS). An allocation method when configured through port-access will replace the user configured method.

The **no** form of this command resets the action to default.

Parameter	Description
usage	Configures the usage-based allocation method.
class	Configures the class-based allocation method.

Usage

If you enable **pd-class-override** for an interface, the **allocate-by** configuration of that interface will be automatically changed to **class**. However, if you change the allocation method to **usage** when **pd-class-override** is still enabled, you will receive an error message stating that "The power allocation method cannot be changed when pd-class-override is enabled."

To remove **pd-class-override**, you can use the **no power-over-ethernet pd-class-override** command . It is important to note that **pd-class-override** requires the allocation method to be set to **class** and is enforced when configured through CLI. However, if you override the allocation method to **usage** via port-access, **pd-class-override** will not be in effect. Therefore, it is recommended that you do not override the allocation method to **usage** through port-access on interfaces configured with **pd-class-override**.

Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet always-on

```
power-over-ethernet always-on <MODULE-ID>
no power-over-ethernet always-on <MODULE-ID>
```

Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The **no** form of this command disables power distribution on soft reboot.

Parameter	Description
<code><MODULE-ID></code>	Module number to apply always-on PoE configuration.

Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet assigned-class

```
power-over-ethernet assigned-class {3 | 4 | 6}  
no power-over-ethernet assigned-class
```

Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the subsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The **no** form of this command resets the action to default.

Examples

Setting PoE assigned class:


```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet pre-std-detect

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. HPE Aruba Networking switches are backward compatible and will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The **no** form of this command resets the action to default.

Examples

Enabling pre-standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling pre-standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet priority

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

Description

Sets PoE priority for an interface. Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet quick-poe

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet quick-poe <MODULE-ID>
```

Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The **no** form of this command disables quick PoE.

Parameter	Description
<MODULE-ID>	Specifies module number for quick PoE configuration .

Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
```

```
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
```

```
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

power-over-ethernet threshold

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.

The **no** form of this command resets the action to default.

Parameter	Description
<code><PERCENTAGE></code>	Excess power consumption trap threshold. Range 1-99.

Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config	Administrators or local user group members with execution rights for this command.

power-over-ethernet trap

```
power-over-ethernet trap  
no power-over-ethernet trap
```

Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The **no** form of this command resets the priority to default PoE priority "low".

Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	config-if	Administrators or local user group members with execution rights for this command.

show lldp local

show lldp local-device [<INTERFACE-ID>]

Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====

Port-ID           : 1/1/10
Port-Desc         : "1/1/10"
Port VLAN ID      : 0

PoE Plus Information

PoE Device Type   : Type 2 PSE
Power Source      : Primary
Power Priority     : low
PSE Allocated Power: 25.0 W
PD Requested Power: 25.0 W
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	

show lldp neighbor

show lldp neighbor [<INTERFACE-ID>]

Description

Displays detailed information about a particular neighbor connected to a particular interface.

Parameter	Description
<INTERFACE-ID>	Specifies an interface. Format: member/slot/port

Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port : 1/1/10
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description : ArubaOS (MODEL: 325), Version HPE_ANW IAP
Neighbor Chassis-ID : 84:d4:7e:ce:5d:68
Neighbor Management-Address : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled :
Neighbor Port-ID : 84:d4:7e:ce:5d:68
Neighbor Port-Desc : eth0
TTL : 120
Neighbor Port VLAN ID :
Neighbor PoEplus information : DOT3
Neighbor Device Type : TYPE2 PD
Neighbor Power Priority : Unkown
Neighbor Power Source : Primary
Neighbor Power Requested : 25.0 W
Neighbor Power Allocated : 0.0 W
Neighbor Power Supported : No
Neighbor Power Enabled : No
Neighbor Power Class : 5
Neighbor Power Paircontrol : No
Neighbor Power Pairs : SIGNAL
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	

show power-over-ethernet

show power-over-ethernet [member <MEMBER-ID>] [brief]

Description

Displays the status information of the full system.

Parameter	Description
<MEMBER-ID>	Displays the detailed status of given member.
<IFNAME>	Display the detailed status of given port.
brief	Display the brief status of all ports or the given port.

Examples

Showing sample output for show power-over-ethernet on standalone box with VSF capability:

```
switch# show power-over-ethernet

System Power Status for member 1

Configured Power Status      : No redundancy
Operational Power Status     : No redundancy
Total Available Power        : 740 W
Total Failover Pwr Avl       : 0 W
Total Redundancy Power       : 0 W
Total Power Drawn            : 0 W +/- 6W
Total Power Reserved         : 0 W
Total Remaining Power        : 740 W
Trap Threshold               : 80 %
Trap Enabled                 : Yes
Always-on PoE Enabled        : 1/1
Quick PoE Enabled            : None

Internal Power
      Total Power
PS    (Watts)    Status
-----
1      0         Absent
2     740        Ok

System Power Status for member 2

Configured Power Status      : No redundancy
Operational Power Status     : No redundancy
Total Available Power        : 600 W
Total Failover Pwr Avl       : 0 W
Total Redundancy Power       : 0 W
Total Power Drawn            : 0 W +/- 6W
Total Power Reserved         : 0 W
```

```

Total Remaining Power      : 600 W
Trap Threshold             : 80 %
Trap Enabled               : Yes
Always-on PoE Enabled     : None
Quick PoE Enabled         : None

```

Internal Power

PS	Total Power (Watts)	Status
1	0	Absent
2	600	Ok

Showing sample output for power-over-ethernet member:

```
switch# show power-over-ethernet member 1
```

System Power Status for member 1

```

Configured Power Status    : No redundancy
Operational Power Status   : No redundancy
Total Available Power      : 740 W
Total Failover Pwr Avl     : 0 W
Total Redundancy Power     : 0 W
Total Power Drawn          : 0 W +/- 6W
Total Power Reserved       : 0 W
Total Remaining Power      : 740 W
Trap Threshold             : 80 %
Trap Enabled               : No
Always-on PoE Enabled     : 1/1
Quick PoE Enabled         : 1/1

```

Internal Power

PS	Total Power (Watts)	Status
1	0	Absent
2	740	Ok

Showing sample output for power-over-ethernet brief in a VSF stack:

```
switch# show power-over-ethernet brief
```

Status and Configuration Information for PoE

Member 1 Power Status

```

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W
Always-on PoE Enabled: 1/1
Quick PoE Enabled: None

```

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2
1/1/2	Yes	Critical	Off	Usage	1.6 W	1.5 W	Delivering*	Single	0	1
1/1/3	Yes	High	Off	Class	54.0 W	25.5 W	Delivering**^	Dual	1/3	3
1/1/4	No	Low	On	Usage	0.0 W	0.0 W	Disabled	None	N/A	N/A

Member 2 Power Status

Available: 600 W Reserved: 0.00 W Remaining: 600 W

Always-on PoE Enabled: None

Quick PoE Enabled: None

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
-----	---	-----	-----	-----	-----	-----	-----	-----	---	----
2/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Searching	None	N/A	N/A
2/1/2	Yes	Critical	Off	Usage	0.0 W	0.0 W	Searching	None	N/A	N/A
2/1/3	Yes	High	Off	Class	0.0 W	0.0 W	Searching	None	N/A	N/A
2/1/4	No	Low	On	Usage	0.0 W	0.0 W	Disabled	None	N/A	N/A

*This port may go down in the event of a PSU failure.

^This port is power demoted due to user config or power availability.

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief
```

Status and Configuration Information for port 1/1/1

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
-----	---	-----	-----	-----	-----	-----	-----	-----	---	----
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2

Showing sample output for power-over-ethernet brief for interface range:

For 6300 Switch series:

```
switch# show power-over-ethernet 1/1/1-1/1/2 brief
```

Status and Configuration Information for port 1/1/1-1/1/2

Member 1Power Status

Available: 370 W Reserved: 55.60 W Remaining: 314.40 W

Always-on PoE Enabled: 1/1

Quick PoE Enabled: None

PoE Port	Pwr En	Power Priority	Pre-std Detect	Alloc Act	PSE Pwr Rsrvd	PD Pwr Draw	PoE Port Status	PD Sign	Cls	Type
-----	---	-----	-----	-----	-----	-----	-----	-----	---	----
1/1/1	Yes	Low	Off	Class	0.0 W	0.0 W	Denied	None	4	2
1/1/2	Yes	Critical	Off	Usage	1.6 W	1.5 W	Delivering*	Single	0	1

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief
```

Module 1/3 is not physically present.

Showing sample output for power-over-ethernet brief for a missing member:

```
switch# show power-over-ethernet member 3 brief
```

```
Member 3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1
```

```
Interface 2/1/1 is not present.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
5420	Operator (>) or Manager (#)	

You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the *AOS-CX Release Notes*.

SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
```

```
snmp-server vrf mgmt
!  
...
```

Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

Device management	Device discovery using SNMPv2C and SNMPv3
	Device dashboards
Monitoring management	Device health attributes (device status/reachability)
	Interface and VLAN management
	Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface.
	Firmware versions
	Displays neighbor devices connected to AOS-CX switches
Configuration management	Device topology
	Partial configuration
Alarm management	Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)
	Syslogs and traps
Report management	Device inventory, interface utilization, and device reachability reports
	Summary report of device model, firmware, and boot loader version

Configuring the AOS-CX switch to be monitored by AirWave

Prerequisites

Aruba AirWave is active on the network.

Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf` command.

```
switch(config)# snmp-server vrf mgmt  
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqBNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as `mgmt`. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as `default` or `mgmt`.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

AirWave commands

logging

```
logging {<IPV4-ADDR> | <IPV6-ADDR> | <FQDN | HOSTNAME>} [ {udp [<PORT-NUM>] }|{tcp
[<PORT-NUM>] | {tls [<PORT-NUM> [auth-mode {certificate|subject-name}] [legacy-tls-
renegotiation]]} [severity <LEVEL>] [vrf <VRF-NAME>] [include-auditable-events]
[filter <FILTER-NAME>] [ rate-limit-burst <BURST> [rate-limit-interval <INTERVAL>] ]
```

```
no logging {<IPV4-ADDR> | <IPV6-ADDR> | <HOSTNAME>}
```

Description

Enables syslog forwarding to a remote syslog server.

The `no` form of this command disables syslog forwarding to a remote syslog server.

Parameter	Description
<code>{<IPv4-ADDR> <IPv6-ADDR> <HOSTNAME>}</code>	Selects the IPv4 address, IPv6 address, or host name of the remote syslog server. Required.
<code>[udp [<PORT-NUM>] tcp [<PORT-NUM>]]</code>	Specifies the UDP port or TCP port of the remote syslog server to receive the forwarded syslog messages.
<code>udp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 514
<code>tcp [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 1470
<code>tls [<PORT-NUM>]</code>	Range: 1 to 65535. Default: 6514
<code>include-auditable-events</code>	Specifies that auditable messages are also logged to the remote syslog server.
<code>severity <LEVEL></code>	Specifies the severity of the syslog messages: ▪ <code>alert</code>: Forwards syslog messages with the severity of <code>alert</code> (6) and <code>emergency</code> (7). ▪ <code>crit</code>: Forwards syslog messages with the severity of <code>critical</code> (5) and above. ▪ <code>debug</code>: Forwards syslog messages with the severity of <code>debug</code> (0) and above. ▪ <code>emerg</code>: Forwards syslog messages with the severity of <code>emergency</code> (7) only. ▪ <code>err</code>: Forwards syslog messages with the severity of <code>err</code> (4) and above ▪ <code>info</code>: Forwards syslog messages with the severity of <code>info</code> (1) and above. Default. ▪ <code>notice</code>: Forwards syslog messages with the severity of <code>notice</code> (2) and above. ▪ <code>warning</code>: Forwards syslog messages with the severity of <code>warning</code> (3) and above.
<code>auth-mode</code>	Specifies the TLS authentication mode used to validate the certificate. ▪ <code>certificate</code>: Validates the peer using trust anchor certificate based authentication. Default. ▪ <code>subject-name</code>: Validates the peer using trust anchor certificates as well as subject-name based authentication.
<code>legacy-tls-renegotiation</code>	Enables the TLS connection with a remote syslog server supporting legacy renegotiation.
<code>vrf <VRF-NAME></code>	Specifies the VRF used to connect to the syslog server. Optional. Default: <code>default</code>

Examples

Enabling the syslog forwarding to remote syslog server 10.0.10.2:

```
switch(config)# logging 10.0.10.2
```

Enabling the syslog forwarding of messages with a severity of `err` (4) and above to TCP port 4242 on remote syslog server 10.0.10.9 with VRF `lab_vrf`:

```
switch(config)# logging 10.0.10.9 tcp 4242 severity err vrf lab_vrf
```

Disabling syslog forwarding to a remote syslog server:

```
switch(config)# no logging
```

Enabling syslog forwarding over TLS to a remote syslog server using subject-name authentication mode:

```
switch(config)# logging example.com tls auth-mode subject name
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server community

```
snmp-server community <STRING>  
no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Parameter	Description
<STRING>	Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server host

```
snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [vrf <VRF-NAME>]
snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
no snmp-server host <IPv4-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [vrf <VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The **no** form of this command removes the specified trap/inform receiver.



Configuring **snmpv3 informs** is not supported.

Parameter	Description
<IPv4-ADDR>	Specifies the IP address of a trap receiver in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255. You can remove leading zeros. For example, the address 192.169.005.100 becomes 192.168.5.100.
trap version <VERSION>	Specifies the trap notification type for SNMPv1 or v2c. Available options are: v1 or v2c.
inform version v2c	Specifies the inform notification type for SNMPv2c.
trap version v3	Specifies the trap notification type for SNMPv3.

Parameter	Description
user <NAME>	Specifies the SNMPv3 user name to be used in the SNMP trap notifications.
community <STRING>	Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: public.
<UDP-PORT>	Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.
vrf <VRF-NAME>	Specifies the name of the VRF on which to send the notifications.

Examples

```

switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000
switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public
port 5000 vrf default
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000
switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
port 5000 vrf default
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community
public port 5000 vrf default

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port
2000

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmp-server vrf

```
snmp-server vrf <VRF-NAME>  
no snmp-server vrf <VRF-NAME>
```

Description

Configures the VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Parameter	Description
<code><VRF-NAME></code>	Specifies the VRF on which the SNMP agent listens for incoming requests. The SNMP agent can listen on either the <code>mgmt</code> or <code>default</code> VRF. If configured for both, the SNMP agent listens on <code>default</code> , which has a higher priority.

Example

```
switch(config)# snmp-server vrf default
```

```
switch(config)# no snmp-server vrf default
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

snmpv3 context

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]  
no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Parameter	Description
<code><NAME></code>	Specifies the name of the context. Range: 1 to 32 printable ASCII

Parameter	Description
	characters, excluding space and question mark (?).
vrf <VRF-NAME>	Specifies the VRF associated with the context. Default: default.
community <STRING>	Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

snmpv3 user

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]
no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context.

The **no** form of this command removes the specified SNMPv3 user.

Parameter	Description
<NAME>	Specifies the SNMPv3 username. Range 1 - 32 printable ASCII characters, excluding space and question mark.
auth <AUTH-PROTOCOL>	Specifies the authentication protocol used to validate user logins. Available options are: md5 or sha.
auth-pass {plaintext ciphertext} <AUTH-PWORD>	Specifies the SNMPv3 user password. Range for plaintext is 8 - 32 printable ASCII characters, excluding space and question

Parameter	Description
	mark. Range for <code>ciphertext</code> is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.
<code>priv <PRIV-PROTOCOL></code>	Specifies the SNMPv3 security protocol (encryption method). Available options are: <code>aes</code> or <code>des</code> .
<code>priv-pass {plaintext ciphertext} <PRIV-PWORD></code>	Specifies the SNMPv3 user privacy passphrase. Range for <code>plaintext</code> is 8 - 32 printable ASCII characters, excluding space and question mark. Range for <code>ciphertext</code> is 1 - 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named **Admin**:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The `snmpv3 user` command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
```

```

!
!Version AOS-CX TL.10.00.0003-8017-gdeb0606~dirty
!
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtFYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmt
!
!
!
!
interface mgmt
    no shutdown
    ip dhcp
vlan 1

```

On switch 2, execute the snmpv3 user command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```

switch1(config)# snmpv3 user Admin auth sha auth-pass
ciphertextAQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtFYbJYCgAAALkGFJVcSp3nZ3o=priv
des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	https://www.hpe.com/us/en/networking/hpe-aruba-networking-support-services.html
AOS-CX Switch Software Documentation Portal	https://arubanetworking.hpe.com/techdocs/AOS-CX/help_portal/Content/home.htm
HPE Aruba Networking Support Portal	https://networkingsupport.hpe.com/home
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-650-750-0350 (Backup—Toll Number)
International telephone	https://www.hpe.com/psnow/doc/a50011948enw

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe-aruba-networking-aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS-CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release: https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP-Q_UL3CskS

HPE Aruba Networking Hardware Documentation and Translations Portal	https://arubanetworking.hpe.com/techdocs/hardware/DocumentationPortal/Content/home.htm
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End-of-Life information	https://networkingsupport.hpe.com/end-of-life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

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(docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.